

# Scientific Computing

Mathematical Foundations for Machine Learning

Course Conspect

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# Course Structure

## Block I: Probability and Statistics

| Lecture | Topic                                                                               |
|---------|-------------------------------------------------------------------------------------|
| 1       | Probability models, conditioning, and random variables                              |
| 2       | Distributions, expectation, and variance                                            |
| 3       | Joint distributions, dependence, and random vectors                                 |
| 4       | Samples, concentration, and limit theorems                                          |
| 5       | Statistical estimation: method of moments, maximum likelihood, and MAP              |
| 6       | Sampling distributions, confidence intervals, and bootstrap                         |
| 7       | Entropy, cross-entropy, KL divergence, Fisher information, and the Cramér–Rao bound |

## Block II: Linear Algebra, Calculus, and Optimization

| Lecture | Topic                                                                          |
|---------|--------------------------------------------------------------------------------|
| 8       | Geometry of data, projections, and least squares                               |
| 9       | Eigenvalues, positive-definite matrices, singular value decomposition, and PCA |
| 10      | Multivariable calculus, matrix calculus, and automatic differentiation         |
| 11      | Optimization for machine learning                                              |

## Block III: Numerical Computing and Autodiff

| Lecture | Topic                                              |
|---------|----------------------------------------------------|
| 12      | NumPy, visualization, and numerical linear algebra |
| 13      | PyTorch, autograd, and the integrated capstone     |

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## Chapter 1

# Probability Models, Conditioning, and Random Variables

Probability starts with a model: a precise description of what may happen, which statements about the outcome are meaningful, and how probabilities are assigned. Conditioning changes the reference population, while a random variable turns outcomes into numerical values whose distribution can be studied.

## 1.1 Probability spaces and events

**Definition 1.1** (Probability space). A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Here  $\Omega$  is the *sample space*, whose elements are outcomes;  $\mathcal{F}$  is a collection of subsets of  $\Omega$ , called events; and  $\mathbb{P}$  assigns a probability to every event.

The event collection  $\mathcal{F}$  is closed under complements and countable unions:

$$A \in \mathcal{F} \implies A^c \in \mathcal{F}, \quad A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

It also contains  $\Omega$ . In a finite model one normally takes  $\mathcal{F} = 2^\Omega$ , the set of all subsets of  $\Omega$ .

For events  $A, B \subseteq \Omega$ :

- $A \cap B$  means that both events occur;
- $A \cup B$  means that at least one occurs;
- $A^c = \Omega \setminus A$  means that  $A$  does not occur;
- $A \setminus B = A \cap B^c$  means that  $A$  occurs but  $B$  does not.

**Definition 1.2** (Probability axioms). For events in  $\mathcal{F}$ , a probability measure satisfies:

$$\begin{aligned} \mathbb{P}(A) &\geq 0, \\ \mathbb{P}(\Omega) &= 1, \\ \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) &= \sum_{i=1}^{\infty} \mathbb{P}(A_i) \quad \text{for pairwise disjoint } A_i. \end{aligned}$$

Several basic rules follow from these axioms:

$$\mathbb{P}(\emptyset) = 0, \quad (1.1)$$

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad (1.2)$$

$$A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B), \quad (1.3)$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \quad (1.4)$$

The last identity is the inclusion–exclusion rule. Adding  $\mathbb{P}(A)$  and  $\mathbb{P}(B)$  counts the overlap twice, so it must be subtracted once.

**Example 1.3** (Two fair coin tosses). Toss a fair coin twice and record the ordered result. Then

$$\Omega = \{HH, HT, TH, TT\}, \quad \mathbb{P}(\{\omega\}) = \frac{1}{4} \quad \text{for every } \omega \in \Omega.$$

Let

$$A = \{\text{first toss is } H\} = \{HH, HT\}, \quad B = \{\text{second toss is } H\} = \{HH, TH\}.$$

Then

$$A \cap B = \{HH\}, \quad A \cup B = \{HH, HT, TH\}, \quad A^c = \{TH, TT\}.$$

Inclusion–exclusion gives

$$\mathbb{P}(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}.$$

For a finite sample space with equally likely outcomes,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

This is a counting shortcut, not an additional axiom: it is valid only after equal likelihood has been justified.

## 1.2 Conditional probability and total probability

**Definition 1.4** (Conditional probability). For  $\mathbb{P}(B) > 0$ , the probability of  $A$  given  $B$  is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (1.5)$$

Conditioning restricts the reference set to  $B$  and renormalizes the remaining probabilities. The event after the vertical bar determines the denominator. In general,  $\mathbb{P}(A \mid B)$  and  $\mathbb{P}(B \mid A)$  are different quantities.

Rearranging (1.5) gives the multiplication rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A). \quad (1.6)$$

**Example 1.5** (At least one head). In the two-toss model, let

$$E = \{\text{at least one head}\} = \{HH, HT, TH\}.$$

Given  $E$ , the probability that the second toss is heads is

$$\mathbb{P}(B \mid E) = \frac{\mathbb{P}(B \cap E)}{\mathbb{P}(E)} = \frac{2/4}{3/4} = \frac{2}{3}.$$

The unconditional probability is  $\mathbb{P}(B) = 1/2$ . Conditioning removes the outcome  $TT$ , in which  $B$  fails.

**Theorem 1.6** (Law of total probability). *Let  $B_1, \dots, B_m$  be a partition of  $\Omega$ : the events are pairwise disjoint and their union is  $\Omega$ . If  $\mathbb{P}(B_i) > 0$ , then*

$$\mathbb{P}(A) = \sum_{i=1}^m \mathbb{P}(A \mid B_i) \mathbb{P}(B_i). \quad (1.7)$$

Indeed, the events  $A \cap B_i$  are disjoint and their union is  $A$ . Add their probabilities and then apply the multiplication rule. For a binary partition  $B, B^c$ ,

$$\mathbb{P}(A) = \mathbb{P}(A \mid B) \mathbb{P}(B) + \mathbb{P}(A \mid B^c) \mathbb{P}(B^c).$$

### 1.3 Independence and mutual exclusivity

**Definition 1.7** (Independence). Events  $A$  and  $B$  are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B). \quad (1.8)$$

If  $\mathbb{P}(B) > 0$ , this is equivalent to  $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ .

Independence means that learning whether one event occurred does not change the probability of the other. In the coin example,

$$\mathbb{P}(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2},$$

so the events “first toss is heads” and “second toss is heads” are independent.

**Definition 1.8** (Mutual exclusivity). Events  $A$  and  $B$  are mutually exclusive, or disjoint, if

$$A \cap B = \emptyset.$$

They cannot occur together.

**Proposition 1.9.** *Two mutually exclusive events with positive probability are not independent.*

*Proof.* Mutual exclusivity gives  $\mathbb{P}(A \cap B) = 0$ , while positive probabilities give  $\mathbb{P}(A) \mathbb{P}(B) > 0$ . Thus (1.8) cannot hold.  $\square$

The distinction is essential:

|                     | Mutual exclusivity      | Independence                                         |
|---------------------|-------------------------|------------------------------------------------------|
| Can both occur?     | No                      | Usually yes                                          |
| Defining condition  | $A \cap B = \emptyset$  | $\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$ |
| Information meaning | One rules out the other | One does not change the other                        |

### 1.4 Bayes' theorem and base rates

**Theorem 1.10** (Bayes' theorem). *Let  $B_1, \dots, B_m$  be a partition of  $\Omega$ . For  $\mathbb{P}(A) > 0$ ,*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j) \mathbb{P}(B_j)}{\sum_{i=1}^m \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)}. \quad (1.9)$$

The numerator is the probability of the route  $B_j$  followed by  $A$ . The denominator adds every mutually exclusive route that can produce  $A$ .

**Example 1.11** (Rare-defect classifier). Let  $D$  mean that an item is defective and  $+$  mean that a classifier produces a positive flag. Assume

$$\mathbb{P}(D) = 0.01, \quad \mathbb{P}(+ | D) = 0.99, \quad \mathbb{P}(+ | D^c) = 0.05.$$

The probability of a positive result is

$$\mathbb{P}(+) = 0.99(0.01) + 0.05(0.99) = 0.0594.$$

Therefore

$$\mathbb{P}(D | +) = \frac{0.99(0.01)}{0.0594} = \frac{1}{6} \approx 0.167.$$

The same calculation can be expressed with expected counts in a population of 10,000 items:

| Actual state    | Positive | Negative | Total  |
|-----------------|----------|----------|--------|
| Defect $D$      | 99       | 1        | 100    |
| No defect $D^c$ | 495      | 9,405    | 9,900  |
| Total           | 594      | 9,406    | 10,000 |

Only 99 of the 594 positive items are defective. The sensitivity  $\mathbb{P}(+ | D) = 0.99$  is not the posterior  $\mathbb{P}(D | +)$ . The low base rate and the false positives among the much larger non-defective group dominate the denominator.

For a Bayesian classifier with classes  $C_1, \dots, C_K$  and observed features  $x$ ,

$$\mathbb{P}(C_k | x) = \frac{p(x | C_k)\mathbb{P}(C_k)}{\sum_{j=1}^K p(x | C_j)\mathbb{P}(C_j)}. \quad (1.10)$$

The factors have distinct roles:

- $\mathbb{P}(C_k)$  is the prior or class base rate;
- $p(x | C_k)$  is the class-conditional likelihood;
- $\mathbb{P}(C_k | x)$  is the posterior after observing  $x$ .

The maximum a posteriori rule is

$$\hat{C}_{\text{MAP}}(x) = \arg \max_k p(x | C_k)\mathbb{P}(C_k).$$

## 1.5 Random variables

**Definition 1.12** (Random variable). A real-valued random variable is a function

$$X : \Omega \longrightarrow \mathbb{R}$$

such that  $\{\omega : X(\omega) \leq x\}$  is an event for every  $x \in \mathbb{R}$ . In finite examples this event condition is automatic.

The sampled outcome is random; the mapping  $X$  is fixed.

**Example 1.13** (Number of heads). For two coin tosses, let  $X$  be the number of heads. Then

| Outcome $\omega$ | $HH$ | $HT$ | $TH$ | $TT$ |
|------------------|------|------|------|------|
| $X(\omega)$      | 2    | 1    | 1    | 0    |

The support  $\{0, 1, 2\}$  is the set of possible values of  $X$ ; it is not the original sample space.

A *discrete* random variable takes values in a finite or countable set. An *absolutely continuous* random variable assigns probabilities to intervals through a density. Mixed random variables, with both point masses and a continuous part, also exist.

## 1.6 PMFs, PDFs, and CDFs

### 1.6.1 Probability mass functions

**Definition 1.14** (PMF). For a discrete random variable  $X$ , its probability mass function is

$$p_X(x) = \mathbb{P}(X = x), \quad p_X(x) \geq 0, \quad \sum_x p_X(x) = 1.$$

For the number of heads in two fair tosses,

$$p_X(0) = \frac{1}{4}, \quad p_X(1) = \frac{1}{2}, \quad p_X(2) = \frac{1}{4}.$$

The probability of an event is obtained by summing the masses of its values; for example,  $\mathbb{P}(X \geq 1) = p_X(1) + p_X(2) = 3/4$ .

### 1.6.2 Cumulative distribution functions

**Definition 1.15** (CDF). For every real-valued random variable  $X$ , the cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Every CDF is nondecreasing and right-continuous, with

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1.$$

It determines interval probabilities through

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a). \quad (1.11)$$

For the number of heads,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{4}, & 0 \leq x < 1, \\ \frac{3}{4}, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

For a discrete variable, the jump at  $x$  equals its point mass:

$$p_X(x) = F_X(x) - F_X(x^-),$$

where  $F_X(x^-)$  denotes the left limit.

### 1.6.3 Probability density functions

**Definition 1.16** (PDF). An absolutely continuous random variable  $Y$  has a probability density function  $f_Y$  satisfying

$$f_Y(y) \geq 0, \quad \int_{-\infty}^{\infty} f_Y(y) dy = 1,$$

and

$$\mathbb{P}(a < Y \leq b) = \int_a^b f_Y(y) dy. \quad (1.12)$$

*Warning.* A density value is not a point probability. For an absolutely continuous random variable,  $\mathbb{P}(Y = y) = 0$  for every single point, and  $f_Y(y)$  may be larger than one. Probability is the area under the density.

The CDF and density are related by

$$F_Y(y) = \int_{-\infty}^y f_Y(t) dt, \quad F'_Y(y) = f_Y(y)$$

at points where the derivative exists.

**Example 1.17** (Uniform distribution on the unit interval). Let

$$f_Y(y) = \begin{cases} 1, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

Thus

$$\mathbb{P}(0.2 < Y \leq 0.7) = \int_{0.2}^{0.7} 1 dy = F_Y(0.7) - F_Y(0.2) = 0.5.$$

## Chapter 2

# Distributions, Expectation, and Variance

## 2.1 Exponential waiting times

Imagine watching a help desk's inbox. Start a stopwatch now and stop it when the next message arrives. The number on the stopwatch is  $T$ , the waiting time in minutes. On one occasion we might wait 0.3 minutes; on another, 7 minutes. We want to answer questions such as: what is the probability that we are still waiting after five minutes?

### 2.1.1 A model of arrivals without a schedule

First build a simple model using one-second intervals. Suppose that in each interval a message arrives with probability  $1/300$ , independently of what happened in earlier intervals. Most seconds contain no message. Occasionally two successful seconds occur close together; occasionally there is a long gap.

In this model, waiting more than five minutes means getting no message in any of 300 consecutive seconds. By independence,

$$\mathbb{P}(T > 5) = \left(1 - \frac{1}{300}\right)^{300} \approx 0.3673.$$

A long wait is therefore quite plausible. The model does not place an arrival at regular five-minute intervals.

Now shorten the time intervals. Write  $\lambda$  for the arrival rate, measured in messages per minute. A small interval of length  $\Delta$  minutes has arrival probability  $\lambda\Delta$ . In our example,  $\lambda = 0.2$ , because  $0.2 \times (1/60) = 1/300$ . The rate describes how frequently messages arrive on average; it is not itself a probability.

Divide a time span of  $t$  minutes into  $n$  equal intervals. In the discrete approximation, each has arrival probability  $\lambda t/n$ . The probability that all of them are empty is

$$\left(1 - \frac{\lambda t}{n}\right)^n.$$

Letting the intervals become arbitrarily short gives

$$\mathbb{P}(T > t) = \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda t}{n}\right)^n = e^{-\lambda t}, \quad t \geq 0. \quad (2.1)$$

This is the probability that *no message has arrived by time  $t$* . The exponential function appears because we multiply the no-arrival probabilities of many short intervals. The resulting continuous waiting time has an *exponential distribution*.

The assumptions describe an inbox with no timetable and no change in activity during the period being studied. They would be unsuitable for notifications sent every hour or an inbox that becomes much busier at lunchtime.

### 2.1.2 From the waiting probability to the density

A message has either arrived by time  $t$ , or it has not. Thus the CDF is

$$F_T(t) = \mathbb{P}(T \leq t) = 1 - e^{-\lambda t}, \quad t \geq 0.$$

Differentiating this CDF, using the relation from Chapter 1, gives the density of the waiting time.

**Definition 2.1** (Exponential distribution). For  $\lambda > 0$ , the notation  $T \sim \text{Exp}(\lambda)$  means

$$f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

The parameter  $\lambda$  is the arrival rate.

Increasing  $\lambda$  makes short waits more likely. For example, at rate 0.2 per minute the probability of waiting more than five minutes is  $e^{-1} \approx 0.3679$ . At rate 0.5, that probability drops to  $e^{-2.5} \approx 0.0821$ . Figure 2.1 shows this comparison directly.

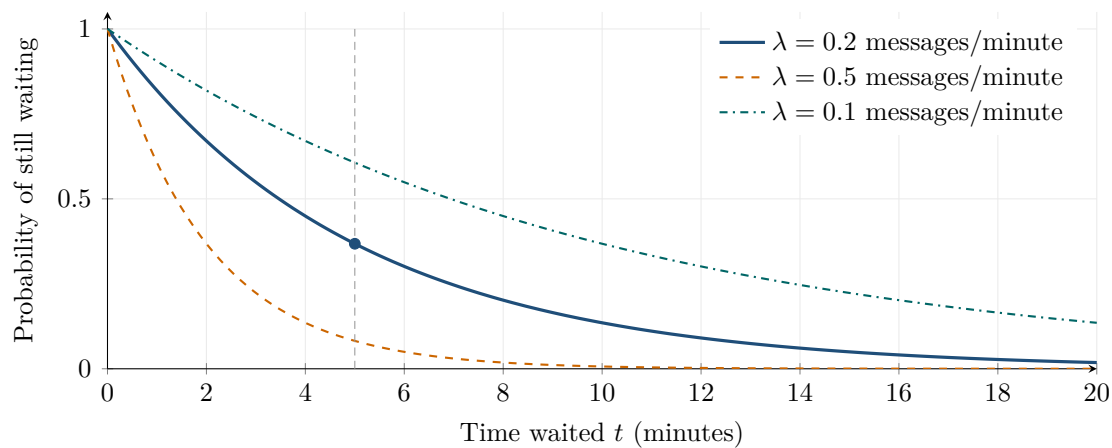


Figure 2.1: The probability that the inbox is still empty after  $t$  minutes. Every curve starts at one: at time zero we have not yet waited. A busier inbox has a larger rate and a faster decrease in the probability of still waiting. The dashed vertical line marks five minutes.

For an interval question, subtract two waiting probabilities. With  $\lambda = 0.2$ ,

$$\mathbb{P}(2 < T \leq 5) = \mathbb{P}(T > 2) - \mathbb{P}(T > 5) = e^{-0.4} - e^{-1} \approx 0.3024.$$

We keep the cases that are still waiting at minute two and remove those that are still waiting at minute five.

A different practical question is how long we must wait to have seen a message in 95% of repetitions. Solve

$$1 - e^{-0.2t} = 0.95 \implies t = -\frac{\log(0.05)}{0.2} \approx 14.98 \text{ minutes.}$$

This cutoff is the 95-th percentile of the waiting time. More generally, the  $q$ -quantile is  $t_q = -\log(1 - q)/\lambda$ , for  $0 < q < 1$ .

### 2.1.3 Does a long wait make an arrival overdue?

Suppose we have already waited ten minutes with no message. In the model, the next short interval has the same arrival chance as it did at the start. Earlier empty intervals do not change that chance. Therefore the probability of waiting another five minutes is still  $e^{-0.2 \cdot 5} \approx 0.3679$ .

We can verify this using conditional probability. For  $s, t \geq 0$ ,

$$\mathbb{P}(T > s + t \mid T > s) = \frac{\mathbb{P}(T > s + t)}{\mathbb{P}(T > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}.$$

The right-hand side is  $\mathbb{P}(T > t)$ . The distribution of the remaining wait is the same as the distribution of the original wait. This property is called *memorylessness*.

A scheduled bus behaves differently: if it comes every ten minutes, waiting nine minutes tells us something about the remaining wait. The exponential model is appropriate only when this kind of schedule, ageing, or change in activity is absent.

## 2.2 Expectation and functions of a random variable

The probability of a long wait answers one question. The average amount of waiting answers another. If we repeat the inbox experiment independently many times and average the recorded times, what value should that average be close to?

Start with a simpler example. Suppose a task takes one minute with probability  $3/4$  and nine minutes with probability  $1/4$ . In many repetitions, about three quarters of the tasks take one minute and one quarter take nine minutes. Their average is close to

$$\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot 9 = 3 \text{ minutes.}$$

This probability-weighted average is the *expectation*. It can be three minutes even though no individual task takes three minutes.

### 2.2.1 Averaging with probabilities

For a discrete random variable, weight each possible value by its probability and add. For a continuous variable, a small interval near  $x$  has probability approximately  $f_X(x)\Delta x$ , so its contribution to the average is approximately  $xf_X(x)\Delta x$ . Summing these contributions over increasingly small intervals gives an integral.

**Definition 2.2** (Expectation). The expectation of  $X$  is

$$\mathbb{E}[X] = \begin{cases} \sum x p_X(x), & X \text{ discrete,} \\ \int_{-\infty}^{\infty} x f_X(x) dx, & X \text{ has a density.} \end{cases}$$

For a finite expectation we require  $\mathbb{E}[|X|] < \infty$ : the sum or integral of absolute contributions must converge.

This condition matters when very large values, although rare, contribute so much that no finite average exists. All the distributions in the main part of this chapter have finite means. The Student extension provides an example where the condition can fail.

For exponential waiting times, the definition gives

$$\mathbb{E}[T] = \int_0^\infty t \lambda e^{-\lambda t} dt.$$

To evaluate it, integrate by parts with  $u = t$  and  $dv = \lambda e^{-\lambda t} dt$ :

$$\mathbb{E}[T] = [-te^{-\lambda t}]_0^\infty + \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda}.$$

At rate 0.2 messages per minute, the mean wait is five minutes. This is an average over repeated waits, not an appointment for the next arrival. We already found that more than a third of waits exceed five minutes.

### 2.2.2 First transform the observation, then average

Sometimes we want to average something other than the waiting time itself. For example, assign a response score

$$g(t) = e^{-at}, \quad a > 0.$$

An immediate response scores one; longer waits receive smaller scores. The parameter  $a$  controls how quickly the score decreases with time.

To find the average score, calculate the score for each possible wait and then weight it by the probability of that wait:

$$\mathbb{E}[g(T)] = \int_0^\infty g(t) f_T(t) dt.$$

For this particular score,

$$\mathbb{E}[e^{-aT}] = \int_0^\infty e^{-at} \lambda e^{-\lambda t} dt = \frac{\lambda}{\lambda + a}.$$

This calculation illustrates the general rule

$$\mathbb{E}[g(X)] = \begin{cases} \sum g(x) p_X(x), & X \text{ discrete,} \\ \int_{-\infty}^{\infty} g(x) f_X(x) dx, & X \text{ has a density.} \end{cases} \quad (2.2)$$

We assume  $\mathbb{E}[|g(X)|] < \infty$ . The rule lets us use the distribution of  $X$  directly, without first deriving the distribution of  $g(X)$ .

Scoring the average wait is a different operation. For  $a = \lambda = 1$ ,

$$\mathbb{E}[e^{-T}] = \frac{1}{2}, \quad e^{-\mathbb{E}[T]} = e^{-1} \approx 0.3679.$$

The nonlinear transformation makes the order of the two operations matter.

Another useful choice is  $g(t) = t^2$ . Large waits receive much more weight after squaring. Again integrating by parts,

$$\mathbb{E}[T^2] = \int_0^\infty t^2 \lambda e^{-\lambda t} dt = \frac{2}{\lambda} \mathbb{E}[T] = \frac{2}{\lambda^2}.$$

We will use this second moment to measure spread.

### 2.2.3 Why expectations add

Suppose completing a job requires a wait  $X$  and a processing time  $Y$ . On each repetition, the total is  $X + Y$ . Averaging the totals gives the average waiting time plus the average processing time. This remains true even if long waits tend to accompany long processing times.

**Theorem 2.3** (Linearity of expectation). *Let  $X_1, \dots, X_n$  have finite expectations. For constants  $a_i$  and  $b$ ,*

$$\mathbb{E}\left[b + \sum_{i=1}^n a_i X_i\right] = b + \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

*Independence is not required.*

For a finite sample space, distributing the sum makes the reason explicit:

$$\sum_{\omega \in \Omega} (X(\omega) + Y(\omega)) \mathbb{P}(\{\omega\}) = \mathbb{E}[X] + \mathbb{E}[Y].$$

The integral version works in the same way.

For instance, if a job has a fixed two-minute setup followed by  $T \sim \text{Exp}(0.2)$ , its mean duration is  $2 + \mathbb{E}[T] = 7$  minutes. Expressing  $T$  in seconds instead multiplies its mean by 60.

Linearity also makes counts easy to analyze. For an event  $A$ , the indicator  $\mathbf{1}_A$  is one if  $A$  occurs and zero otherwise. Its expectation is

$$\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \mathbb{P}(A^c) = \mathbb{P}(A).$$

Consequently, the expected number of events occurring among  $A_1, \dots, A_n$  is

$$\mathbb{E}\left[\sum_{i=1}^n \mathbf{1}_{A_i}\right] = \sum_{i=1}^n \mathbb{P}(A_i).$$

No independence assumption is needed for this mean. Dependence will matter when we ask about the count's spread.

## 2.3 Variance and standard deviation

A constant five-minute wait and an exponential wait with mean five minutes have the same expectation. Their behavior is very different: the first is perfectly predictable, while the second sometimes ends almost immediately and sometimes takes much longer. We need a measure of this variation around the mean.

If  $\mu = \mathbb{E}[X]$ , the deviation of an observation is  $X - \mu$ . Averaging the signed deviations gives zero:  $\mathbb{E}[X - \mu] = 0$ . Positive and negative deviations cancel. Squaring them prevents that cancellation and gives more weight to large departures from the mean.

**Definition 2.4** (Variance and standard deviation). For  $X$  with a finite second moment,

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2], \quad \mu = \mathbb{E}[X], \quad \text{sd}(X) = \sqrt{\text{Var}(X)}.$$

Variance is the average squared deviation. If waiting time is in minutes, variance is in minutes squared. Taking the square root returns to minutes; this is why standard deviation is useful when describing the size of typical fluctuations. It is not a bound on individual observations.

Expanding the square gives a convenient way to calculate variance:

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2.\end{aligned}\tag{2.3}$$

For exponential waiting times, we have already computed both terms:

$$\text{Var}(T) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}, \quad \text{sd}(T) = \frac{1}{\lambda}.$$

The inbox with rate 0.2 has both mean and standard deviation equal to five minutes. A constant five-minute wait instead has variance zero.

### 2.3.1 Changing the origin or the units

Adding a fixed delay  $b$  moves every observation and its mean by the same amount. The deviations from the mean stay unchanged. Multiplying times by  $a$  multiplies every deviation by  $a$ , so their squares are multiplied by  $a^2$ . Therefore

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

For example, converting minutes to seconds multiplies the mean and standard deviation by 60, and the variance by  $60^2$ . Variance is not linear.

### 2.3.2 Adding two variable delays

For a job with waiting time  $X$  and processing time  $Y$ , let  $U = X - \mathbb{E}[X]$  and  $V = Y - \mathbb{E}[Y]$ . The deviation of the total duration is  $U + V$ . Squaring gives

$$(U + V)^2 = U^2 + V^2 + 2UV.$$

The first two terms describe the separate spreads. The last term tells us whether the two delays fluctuate together. If both are above their means, or both are below, their product  $UV$  is positive. If one is above its mean while the other is below, the product is negative.

The average of this product is called the *covariance*:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Taking expectations in the squared-sum identity gives

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).\tag{2.4}$$

For independent variables, expectations of products factor. Thus  $\mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] = 0$ , and the covariance term vanishes. For independent variables with finite second moments, this extends to

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i).$$

**Example 2.5** (Three different ways to combine fluctuations). Let  $X, Y$  each have variance one. If they are independent, the variance of  $X + Y$  is two. If  $Y = X$ , the fluctuations reinforce each other and  $\text{Var}(X + Y) = \text{Var}(2X) = 4$ . If  $Y = -X$ , they cancel exactly and the sum has variance zero.

Zero covariance is enough for variances to add, even without full independence. We will study that distinction further when considering joint distributions. For the sums used below, independence will be an explicit part of the model.

## 2.4 Poisson counts

Return to the inbox, but change the question. Instead of timing the next message, watch for ten minutes and count how many messages arrive. A waiting time can be 2.4 minutes; a message count must be an integer such as zero, one, or two. The same arrivals produce two different kinds of random variables.

Write  $N(t)$  for the number of messages arriving between time zero and time  $t$ . Figure 2.2 shows one possible record. At minute six we have counted two messages; at minute ten we have counted three.

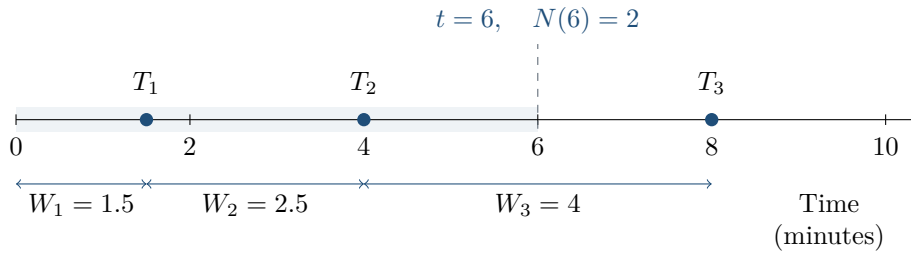


Figure 2.2: One possible sequence of arrivals, at 1.5, 4, and 8 minutes. The gaps  $W_1, W_2, W_3$  are waiting times; the arrival times  $T_k$  are cumulative sums of those gaps. By the marked time  $t = 6$ , two messages have arrived, so  $N(6) = 2$ .

### 2.4.1 Counting successful short intervals

We can use the same short-interval model as before. Divide  $t$  minutes into  $n$  tiny intervals, each independently containing a message with probability  $p_n = \lambda t/n$ . To get exactly  $k$  messages, exactly  $k$  intervals must contain a message and the others must be empty.

For a particular choice of those  $k$  intervals, independence gives probability  $p_n^k (1 - p_n)^{n-k}$ . The number of ways to choose the intervals is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

For example, if there are three intervals and exactly one message, it can be in the first, second, or third interval: three possibilities. Here  $n! = 1 \cdot 2 \cdots n$ , with  $0! = 1$ . Adding the equally probable possibilities gives

$$\binom{n}{k} \left( \frac{\lambda t}{n} \right)^k \left( 1 - \frac{\lambda t}{n} \right)^{n-k}.$$

Put  $m = \lambda t$ . For fixed  $k$ , as  $n$  increases,

$$\frac{\binom{n}{k}}{n^k} \longrightarrow \frac{1}{k!}, \quad \left( 1 - \frac{m}{n} \right)^{n-k} \longrightarrow e^{-m}.$$

Thus the continuous-time count has probability  $e^{-m} m^k / k!$  of equalling  $k$ . This is the *Poisson distribution*.

**Definition 2.6** (Poisson distribution). For  $m > 0$ , a count  $N \sim \text{Poisson}(m)$  has PMF

$$\mathbb{P}(N = k) = e^{-m} \frac{m^k}{k!}, \quad k = 0, 1, 2, \dots$$

For our inbox observed for  $t$  minutes,  $N(t) \sim \text{Poisson}(\lambda t)$ .

The rate  $\lambda$  describes arrivals per minute;  $m = \lambda t$  describes the expected number over the whole observation window. At rate 0.2 per minute, a ten-minute window has  $m = 2$ , while a thirty-minute window has  $m = 6$ . Neither count is guaranteed to equal its mean.

**Example 2.7** (No messages or several messages). For the ten-minute count with  $m = 2$ ,

$$\mathbb{P}(N(10) = 0) = e^{-2} \approx 0.1353,$$

and

$$\mathbb{P}(N(10) \geq 2) = 1 - \mathbb{P}(N(10) = 0) - \mathbb{P}(N(10) = 1) = 1 - e^{-2}(1 + 2) \approx 0.5940.$$

No messages in ten minutes is exactly the event that the first waiting time exceeds ten minutes. The Poisson and Exponential calculations agree:  $\mathbb{P}(N(10) = 0) = \mathbb{P}(T > 10) = e^{-2}$ .

### 2.4.2 Mean and variance of the count

To check normalization, sum the PMF and use the exponential series:

$$\sum_{k=0}^{\infty} e^{-m} \frac{m^k}{k!} = e^{-m} e^m = 1.$$

For the mean, the factor  $k$  cancels one factor in  $k!$ . Reindexing the remaining sum with  $j = k - 1$  gives

$$\mathbb{E}[N] = \sum_{k=1}^{\infty} k e^{-m} \frac{m^k}{k!} = m \sum_{j=0}^{\infty} e^{-m} \frac{m^j}{j!} = m.$$

For the variance we need  $\mathbb{E}[N^2]$ . It is easier to first calculate  $\mathbb{E}[N(N - 1)]$ , since  $k(k - 1)$  cancels two factors in  $k!$ :

$$\mathbb{E}[N(N - 1)] = m^2 \sum_{j=0}^{\infty} e^{-m} \frac{m^j}{j!} = m^2.$$

Now use  $N^2 = N(N - 1) + N$ :

$$\mathbb{E}[N^2] = m^2 + m, \quad \text{Var}(N) = m.$$

For ten minutes, the mean and variance are both two, and the standard deviation is  $\sqrt{2}$  messages.

The equality of mean and variance will later help us recognize when a single Poisson model misses part of the variation in a population.

### 2.4.3 Combining independent arrival streams

Suppose two independent inboxes receive messages at rates  $\lambda_1$  and  $\lambda_2$ . Counting all their messages over the same interval should add the expected counts. In fact, the entire distribution remains Poisson:

$$N_1 + N_2 \sim \text{Poisson}(m_1 + m_2), \quad m_i = \lambda_i t.$$

To see why, split a total of  $k$  messages into  $j$  from the first inbox and  $k - j$  from the second. Independence lets us multiply their probabilities; summing over  $j$  and using the binomial theorem gives

$$\mathbb{P}(N_1 + N_2 = k) = \frac{e^{-(m_1+m_2)}}{k!} \sum_{j=0}^k \binom{k}{j} m_1^j m_2^{k-j} = \frac{e^{-(m_1+m_2)} (m_1 + m_2)^k}{k!}.$$

In the short-interval model, counts in disjoint time windows are also independent. If arrivals instead trigger further arrivals or the rate changes with the time of day, these assumptions need to be reconsidered.

## 2.5 Gamma waiting times and the event-count connection

Suppose the help desk collects three messages before starting a batch. Now we need the time until the third message, rather than the time until the next one or the number received by a fixed deadline.

Let  $W_1$  be the wait from time zero to the first message,  $W_2$  the gap from the first to the second, and so on. In our arrival model these gaps are independent and each follows  $\text{Exp}(\lambda)$ . The  $k$ -th arrival time is

$$T_k = W_1 + \cdots + W_k.$$

For example, the gaps in Figure 2.2 are 1.5, 2.5, and 4 minutes, so  $T_3 = 8$  minutes.

Before deriving a new density, we can already find the mean and variance:

$$\mathbb{E}[T_k] = \frac{k}{\lambda}, \quad \text{Var}(T_k) = \frac{k}{\lambda^2}.$$

Expectation adds by linearity; variance adds because the gaps are independent. At rate 0.2, waiting for three messages takes 15 minutes on average, with variance 75 minutes squared.

### 2.5.1 The distribution of a sum of waiting times

Consider two messages first. A total wait near five minutes could consist of a first gap near one minute and a second near four, or a first near two and a second near three, and so on. To find the density of the total, we must account for every possible split.

If the first gap is  $u$ , the second must be near  $t - u$ . Independence gives a product of densities for these two contributions. Integrating over  $0 < u < t$  adds all the possible splits:

$$\begin{aligned} f_{T_2}(t) &= \int_0^t f_{W_1}(u) f_{W_2}(t-u) du \\ &= \int_0^t \lambda e^{-\lambda u} \lambda e^{-\lambda(t-u)} du = \lambda^2 t e^{-\lambda t}, \quad t > 0. \end{aligned}$$

This way of finding the density of a sum is called *convolution*.

Repeating the same step for  $k$  independent gaps gives

$$f_{T_k}(t) = \frac{\lambda^k}{(k-1)!} t^{k-1} e^{-\lambda t}, \quad t > 0.$$

For  $k = 1$ , this is the exponential density. For  $k > 1$ , the density starts near zero: fitting several arrivals into a very short time is unlikely. As  $t$  increases, enough time becomes available for all  $k$  arrivals. Eventually the density decreases because exceptionally long total waits are also unlikely.

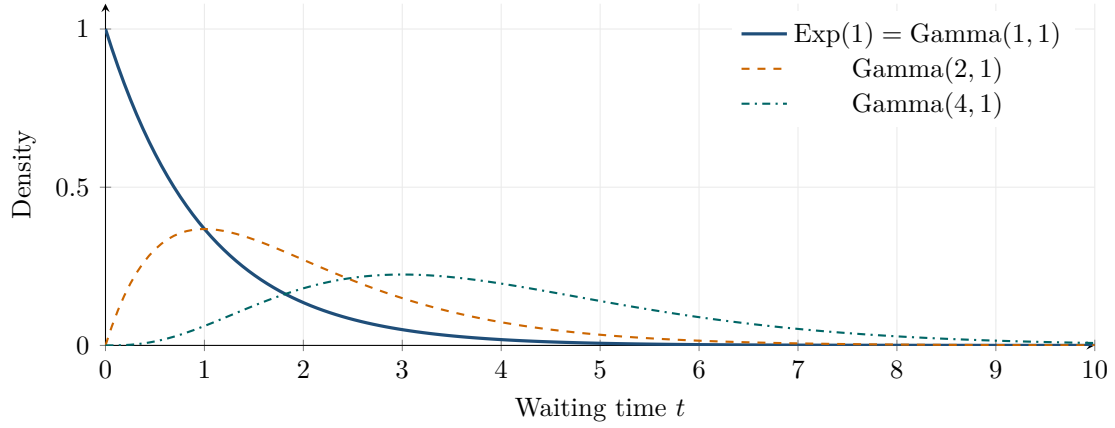


Figure 2.3: Time until the first, second, or fourth arrival, at rate one. One arrival can occur almost immediately. Several arrivals require a sum of waiting times, moving the typical total wait to the right. The means are 1, 2, 4; the standard deviations are 1,  $\sqrt{2}$ , 2.

### 2.5.2 From an integer number of arrivals to a shape parameter

The density above gives a useful family of positive, often skewed distributions. We can extend its shape beyond integer values of  $k$ . The only obstacle is the factorial  $(k-1)!$ . Its continuous extension is the gamma function:

$$\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} du, \quad \alpha > 0.$$

Integration by parts gives  $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ . Since  $\Gamma(1) = 1$ , this implies  $\Gamma(k) = (k-1)!$  for positive integers  $k$ .

Replacing  $k$  by a positive real number  $\alpha$  and the factorial by  $\Gamma(\alpha)$  gives the *Gamma distribution*. We write its rate as  $\beta$  so that  $\lambda$  can continue to denote the particular inbox rate.

**Definition 2.8** (Gamma distribution). For shape  $\alpha > 0$  and rate  $\beta > 0$ ,  $X \sim \text{Gamma}(\alpha, \beta)$  has density

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0,$$

and zero density for  $x < 0$ .

The substitution  $u = \beta x$  shows that its integral is one. The case  $\alpha = 1$  is Exponential; integer  $\alpha = k$  describes waiting for  $k$  arrivals at rate  $\beta$ . For noninteger shape, the density still makes sense, but there is no literal fractional-number-of-arrivals interpretation.

We can calculate the moments for every positive shape by inserting  $x$  or  $x^2$  into the density integral:

$$\begin{aligned} \mathbb{E}[X] &= \frac{\Gamma(\alpha+1)}{\beta\Gamma(\alpha)} = \frac{\alpha}{\beta}, \\ \mathbb{E}[X^2] &= \frac{\Gamma(\alpha+2)}{\beta^2\Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\beta^2}. \end{aligned}$$

Subtracting the squared mean gives

$$\text{Var}(X) = \frac{\alpha}{\beta^2}.$$

Increasing the rate  $\beta$  shortens the waits. Increasing the shape  $\alpha$  at fixed rate increases the mean, while the spread relative to that mean decreases:  $\text{sd}(X)/\mathbb{E}[X] = 1/\sqrt{\alpha}$ .

Throughout these notes the second Gamma parameter is a rate. Another common convention uses a scale  $\theta = 1/\beta$ , measured in the same units as  $X$ . In that convention the mean and variance are  $\alpha\theta$  and  $\alpha\theta^2$ . Checking which parameterization is used prevents a reciprocal error.

### 2.5.3 A deadline question can be answered by counting

To have received the third message by minute ten means to have counted at least three messages by minute ten. These are the same event, expressed using different random variables:

$$\{T_k \leq t\} = \{N(t) \geq k\}, \quad k \geq 1. \quad (2.5)$$

Therefore, instead of integrating a Gamma density, we can add Poisson probabilities:

$$\mathbb{P}(T_k \leq t) = 1 - \mathbb{P}(N(t) < k) = 1 - e^{-\lambda t} \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!}.$$

For rate 0.2, a ten-minute window has expected count two, so

$$\mathbb{P}(T_3 \leq 10) = 1 - e^{-2}(1 + 2 + 2^2/2) \approx 0.3233.$$

Receiving exactly  $k$  messages also has a waiting-time description: the  $k$ -th has arrived, but the  $(k+1)$ -th has not. For  $k \geq 1$ ,  $\mathbb{P}(N(t) = k) = \mathbb{P}(T_k \leq t) - \mathbb{P}(T_{k+1} \leq t)$ . For  $k = 0$ , it is simply  $\mathbb{P}(T_1 > t)$ .

## 2.6 Bernoulli and Binomial distributions

The inbox model asked whether a message arrived during each small time interval. Many other observations are also binary: a link was clicked or it was not, a transmission succeeded or failed, an item passed an inspection or failed it. We now describe these trials without reference to time.

### 2.6.1 One binary outcome

Code a click as one and no click as zero. If the click probability is  $p$ , then

$$\mathbb{P}(B = 1) = p, \quad \mathbb{P}(B = 0) = 1 - p, \quad 0 \leq p \leq 1.$$

This is a *Bernoulli distribution*, written  $B \sim \text{Bernoulli}(p)$ . The same construction applies to the indicator of any event.

For a binary variable, squaring changes neither possible value:  $0^2 = 0$ ,  $1^2 = 1$ . Thus

$$\mathbb{E}[B] = p, \quad \mathbb{E}[B^2] = p, \quad \text{Var}(B) = p(1 - p).$$

At  $p = 1/2$ , neither outcome is favored and the variance is largest. At  $p = 0$  or  $p = 1$ , every observation is the same and variance is zero.

### 2.6.2 A fixed number of independent trials

Show a link to  $n$  people. Assume their clicks are independent and each has the same probability  $p$ . If  $B_i$  indicates the  $i$ -th click, then the total number of clicks is

$$S_n = B_1 + \cdots + B_n.$$

It follows a *Binomial distribution*,  $S_n \sim \text{Binomial}(n, p)$ , with

$$\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n.$$

The counting argument is the one used for the short arrival intervals: choose the  $k$  successful trials, multiply their success and failure probabilities, and add over all choices. The displayed expression is read directly for  $0 < p < 1$ ; at  $p = 0$  or  $p = 1$  the count is fixed.

We do not need to sum this PMF to obtain its moments. Since the count is a sum of independent Bernoulli variables,

$$\mathbb{E}[S_n] = np, \quad \text{Var}(S_n) = np(1-p).$$

For  $n \geq 1$ , the observed click proportion is  $\hat{p} = S_n/n$ . The scaling rules for mean and variance give

$$\mathbb{E}[\hat{p}] = p, \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

With more independent observations, the proportion fluctuates less. For example, multiplying  $n$  by four halves its standard deviation.

**Example 2.9** (A small click probability). For 100 independent impressions with  $p = 0.03$ , the count has mean 3 and variance 2.91. The chance of no clicks is

$$\mathbb{P}(S_{100} = 0) = (1 - 0.03)^{100} \approx 0.0476.$$

The common- $p$  assumption treats all impressions alike. We will revisit it when different users have different propensities to click.

### 2.6.3 Why rare Binomial successes look Poisson

The short-interval model contained many independent trials, each with a tiny success probability. A Binomial count behaves similarly whenever  $n$  is large and  $p$  is small, with  $np = m$  held fixed. For any fixed  $k$ ,

$$\binom{n}{k} \left(\frac{m}{n}\right)^k \left(1 - \frac{m}{n}\right)^{n-k} \longrightarrow e^{-m} \frac{m^k}{k!}.$$

This is the Poisson limit derived earlier. Its moments agree with the Binomial moments in the limit:

$$np = m, \quad np(1-p) \longrightarrow m.$$

For the click example, the approximation is  $\text{Poisson}(3)$ , which gives no-click probability  $e^{-3} \approx 0.0498$ , close to the exact 0.0476.

## 2.7 Geometric waiting times

Now show the link repeatedly until the first click, instead of fixing the number of impressions in advance. Let  $G$  be the number of impressions needed, including the successful one. For example, the sequence

no click,   no click,   click

gives  $G = 3$ .

Assume independent trials with the same success probability  $0 < p \leq 1$ . For  $0 < p < 1$ , the event  $G = k$  requires  $k - 1$  failures followed by one success:

$$\mathbb{P}(G = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

This is the *Geometric distribution*,  $G \sim \text{Geometric}(p)$ . If  $p = 1$ , the first trial always succeeds and  $G = 1$ .

### 2.7.1 Waiting longer means accumulating failures

The event  $G > k$  means that all of the first  $k$  trials failed. Hence

$$\mathbb{P}(G > k) = (1 - p)^k, \quad k = 0, 1, \dots$$

With  $p = 0.1$ , the chance of still waiting for a success after twenty trials is  $0.9^{20} \approx 0.1216$ .

This model has the same memoryless logic as exponential waiting. Once the first  $m$  trials have failed, the next trials still have the original success probability. Formally, for  $0 < p < 1$ ,

$$\mathbb{P}(G > m + k \mid G > m) = \frac{(1 - p)^{m+k}}{(1 - p)^m} = (1 - p)^k.$$

An example is a scrolling session in which a user independently stops after each video with probability  $p$ . The number of videos watched is Geometric. If fatigue makes stopping more likely after each video, the constant- $p$  assumption no longer holds.

### 2.7.2 Mean and variance

To compute the mean, weight each possible stopping trial by its probability. Write  $q = 1 - p$ . The geometric series

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1 - q}, \quad |q| < 1,$$

can be differentiated to obtain the factor  $k$  needed in the mean:

$$\sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1 - q)^2}.$$

Multiplying by  $p$  gives  $\mathbb{E}[G] = 1/p$ . A second differentiation gives

$$\mathbb{E}[G(G - 1)] = p \sum_{k=2}^{\infty} k(k - 1)q^{k-1} = \frac{2(1 - p)}{p^2}.$$

As in the Poisson calculation, use  $G^2 = G(G - 1) + G$  to get

$$\text{Var}(G) = \frac{1 - p}{p^2}.$$

At  $p = 0.1$ , the mean is ten trials and the variance is ninety.

Some sources count failures before success instead. That variable is  $H = G - 1$ , so its mean is  $(1 - p)/p$ , its variance is unchanged, and it can equal zero. In these notes, Geometric always counts the successful trial too.

### 2.7.3 Turning a trial count into a time

Return to the inbox approximation, where one trial corresponds to a short interval of duration  $\Delta$  minutes. A count of  $G$  intervals represents a waiting time of  $\Delta G$  minutes. Set  $p = \lambda\Delta$ , just as in the first section.

By time  $t$ , we have completed  $\lfloor t/\Delta \rfloor$  intervals; the floor symbol means the integer number of whole intervals. No arrival by that time has probability

$$\mathbb{P}(\Delta G > t) = (1 - \lambda\Delta)^{\lfloor t/\Delta \rfloor} \longrightarrow e^{-\lambda t} \quad (\Delta \downarrow 0).$$

The Geometric waiting count therefore becomes an Exponential waiting time as the time intervals shrink. The moments show the same connection:

$$\mathbb{E}[\Delta G] = \frac{1}{\lambda}, \quad \text{Var}(\Delta G) = \frac{1 - \lambda\Delta}{\lambda^2} \longrightarrow \frac{1}{\lambda^2}.$$

The time-unit conversion is what connects the two distributions: Geometric counts trials; Exponential measures elapsed time.

## 2.8 The Gaussian distribution

The waiting times studied so far are nonnegative and usually skewed: a wait can be much longer than average, but cannot extend below zero. Measurement errors pose a different problem. A sensor measuring a fixed quantity may sometimes read too high and sometimes too low. Several small disturbances can add together, with positive and negative contributions partly cancelling.

A useful model for such errors has a central peak and decreases symmetrically on either side. This is the *Gaussian*, also called the *normal*, distribution. It is often a useful approximation when many small independent disturbances contribute and no one disturbance dominates. The connection between sums and Gaussian approximations will be developed further in the chapter on limit theorems.

### 2.8.1 Center and width

Let  $\mu$  describe the center of the errors and  $\sigma > 0$  their scale. The center can be nonzero if the sensor is systematically biased. The density is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right], \quad x \in \mathbb{R}.$$

We write  $X \sim \mathcal{N}(\mu, \sigma^2)$ .

The expression  $(x - \mu)/\sigma$  measures how far  $x$  lies from the center relative to the width of the curve. Squaring it treats positive and negative deviations equally. The negative exponential

makes values far from the center less likely. Increasing  $\sigma$  widens the curve; the factor  $1/\sigma$  reduces its height to preserve total area one.

The simplest member is  $Z \sim \mathcal{N}(0, 1)$ , with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

The constant  $\sqrt{2\pi}$  is the value of  $\int_{-\infty}^{\infty} e^{-z^2/2} dz$ , so this density integrates to one. Every other Gaussian can be obtained by shifting and scaling:

$$X = \mu + \sigma Z.$$

### 2.8.2 Why the parameters are the mean and standard deviation

Symmetry of  $\phi$  means that the contributions of  $z$  and  $-z$  to the mean cancel, giving  $\mathbb{E}[Z] = 0$ . For the second moment, differentiating the density gives  $\phi'(z) = -z\phi(z)$ . This turns the moment integral into an integration-by-parts calculation:

$$\begin{aligned} \mathbb{E}[Z^2] &= \int_{-\infty}^{\infty} z^2 \phi(z) dz = - \int_{-\infty}^{\infty} z \phi'(z) dz \\ &= [-z\phi(z)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z) dz = 1. \end{aligned}$$

Thus  $\text{Var}(Z) = 1$ . Applying the shift and scale rules to  $X = \mu + \sigma Z$  gives

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2, \quad \text{sd}(X) = \sigma.$$

This justifies the parameters' interpretation. In  $\mathcal{N}(\mu, \sigma^2)$ , the second argument is the variance.

### 2.8.3 Expressing an error in standard-deviation units

For a sensor with mean error zero, an error of 0.4 units may be large or small relative to its usual variation. If the standard deviation is 0.2, that error is two standard deviations above the mean; if it is 2, it is only 0.2 standard deviations above the mean.

The transformation

$$Z = \frac{X - \mu}{\sigma}$$

subtracts the center and expresses the deviation in standard-deviation units. This is *standardization*. For a Gaussian  $X$ , the result always follows  $\mathcal{N}(0, 1)$ . We can therefore answer interval questions for any Gaussian using a single CDF:

$$\Phi(z) = \mathbb{P}(Z \leq z) = \int_{-\infty}^z \phi(u) du.$$

The endpoints of the original interval are transformed in the same way:

$$\mathbb{P}(a < X \leq b) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

The integral defining  $\Phi$  is evaluated numerically. Symmetry gives  $\Phi(-z) = 1 - \Phi(z)$ .

**Example 2.10** (How often does a sensor make a large error?). Suppose the error is  $E \sim \mathcal{N}(0, 0.2^2)$ . An absolute error larger than 0.4 is an error more than two standard deviations from zero. The two symmetric tails give

$$\mathbb{P}(|E| > 0.4) = \mathbb{P}(|Z| > 2) = 2[1 - \Phi(2)] \approx 0.0455.$$

Under this model, about 4.55% of repeated readings have such an error. Likewise,

$$\mathbb{P}(|Z| \leq 1) \approx 0.6827, \quad \mathbb{P}(|Z| \leq 1.96) \approx 0.95.$$

The central 95% interval for the sensor error is therefore

$$[-1.96(0.2), 1.96(0.2)] = [-0.392, 0.392].$$

### 2.8.4 Adding errors and connecting back to earlier distributions

Changing units or adding a fixed bias preserves the Gaussian shape:

$$aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2), \quad a \neq 0.$$

For  $a = 0$ , the result is simply the constant  $b$ .

Independent Gaussian errors also add to another Gaussian. If  $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$  and  $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$  are independent, then

$$X + Y \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

The convolution method used for Gamma waiting times gives this density. The means and variances agree with the addition rules we already know.

For example, two independent centered errors with standard deviations 0.2 and 0.3 produce a total error with variance  $0.2^2 + 0.3^2 = 0.13$ , and standard deviation  $\sqrt{0.13} \approx 0.361$ . Standard deviations do not simply add.

A sum can also be approximately Gaussian even when its individual terms are not Gaussian. We have already met two such sums: Binomial counts add Bernoulli outcomes, and integer-shape Gamma waits add exponential gaps. For sufficiently large parameters, useful approximations are

$$\begin{aligned} \text{Binomial}(n, p) &\approx \mathcal{N}(np, np(1-p)), \\ \text{Gamma}(k, \lambda) &\approx \mathcal{N}(k/\lambda, k/\lambda^2), \\ \text{Poisson}(m) &\approx \mathcal{N}(m, m). \end{aligned}$$

Each Gaussian uses the mean and variance of the distribution being approximated. For the Binomial, both the expected successes  $np$  and failures  $n(1-p)$  should be large; for Gamma and Poisson, the approximations improve as  $k$  and  $m$  grow.

The Gaussian allows negative values, while counts and waiting times do not. Thus approximating a very small count or a strongly skewed waiting time by a Gaussian can assign substantial probability to impossible outcomes. The original distributions remain the appropriate starting point in those cases.

## 2.9 Mixtures and heterogeneity

Suppose two sensors measure the same reference quantity. One tends to read two units too low; the other tends to read two units too high. Within either sensor, the random error has standard deviation one.

Choose a sensor with equal probability for each reading. If we combine all readings without keeping track of which sensor produced them, what distribution do the errors follow? Averaging the two sensor biases gives zero, but that does not mean either sensor usually makes an error near zero. Most errors are still near  $-2$  or  $2$ .

### 2.9.1 Choose a source, then draw an observation

Let  $Z$  record which sensor is selected:

$$\mathbb{P}(Z = 1) = \mathbb{P}(Z = 2) = \frac{1}{2}.$$

Given the chosen sensor, the error distribution is

$$X \mid Z = 1 \sim \mathcal{N}(-2, 1), \quad X \mid Z = 2 \sim \mathcal{N}(2, 1).$$

The notation  $X \mid Z = k$  means that we are considering only readings from sensor  $k$ .

Write the two Gaussian densities as  $f_1$  and  $f_2$ . For any interval, half the readings come from each sensor, so the law of total probability gives half of each sensor's interval probability. The density is therefore

$$f_X(x) = \frac{1}{2}f_1(x) + \frac{1}{2}f_2(x).$$

This is a *mixture distribution*. Its two peaks reflect the two sources of readings.

The same construction works with  $K$  sources chosen with probabilities  $\pi_1, \dots, \pi_K$ :

$$f_X(x) = \sum_{k=1}^K \pi_k f_k(x), \quad \pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1. \quad (2.6)$$

The  $f_k$  are called the mixture's *components*. For discrete observations, use component PMFs in the same formula. If the source label is not recorded, it is an unobserved, or *latent*, variable.

### 2.9.2 Where the pooled variance comes from

In the two-sensor example the overall mean is

$$\mathbb{E}[X] = \frac{1}{2}(-2) + \frac{1}{2}(2) = 0.$$

Within either sensor, the second moment equals its variance plus the square of its mean:

$$\mathbb{E}[X^2 \mid Z = 1] = 1 + (-2)^2 = 5, \quad \mathbb{E}[X^2 \mid Z = 2] = 1 + 2^2 = 5.$$

Averaging these gives  $\mathbb{E}[X^2] = 5$ . Hence the pooled variance is  $5 - 0^2 = 5$ , even though each sensor separately has variance one.

There are two sources of spread: random fluctuation within a sensor, and the difference between the two sensors' centers. To separate them in general, let component  $k$  have mean  $\mu_k$  and variance  $\sigma_k^2$ . Then

$$\mu = \mathbb{E}[X] = \sum_k \pi_k \mu_k, \quad \mathbb{E}[X^2] = \sum_k \pi_k (\sigma_k^2 + \mu_k^2).$$

Subtracting  $\mu^2$  and regrouping the terms gives

$$\boxed{\text{Var}(X) = \underbrace{\sum_k \pi_k \sigma_k^2}_{\text{variation within sources}} + \underbrace{\sum_k \pi_k (\mu_k - \mu)^2}_{\text{variation between sources}}.} \quad (2.7)$$

For the sensors, these contributions are 1 and 4, respectively.

The inner averages can be written as conditional expectations. The quantity  $\mathbb{E}[X | Z]$  equals  $\mu_k$  when source  $k$  is selected; it varies because the selected source varies. Similarly,  $\text{Var}(X | Z)$  is the variance inside the selected source. The outer expectation averages over the choice of source. With this notation, the two identities become

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Z]], \quad (2.8)$$

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Z)] + \text{Var}(\mathbb{E}[X | Z]). \quad (2.9)$$

These are the *laws of total expectation and total variance*. We have derived them for a finite mixture with finite second moments.

### 2.9.3 The same mean and variance can hide different shapes

A single Gaussian  $\mathcal{N}(0, 5)$  has the same mean and variance as the two-sensor mixture. Its most likely region is near zero, whereas the mixture has two peaks near the sensor biases (Figure 2.4).

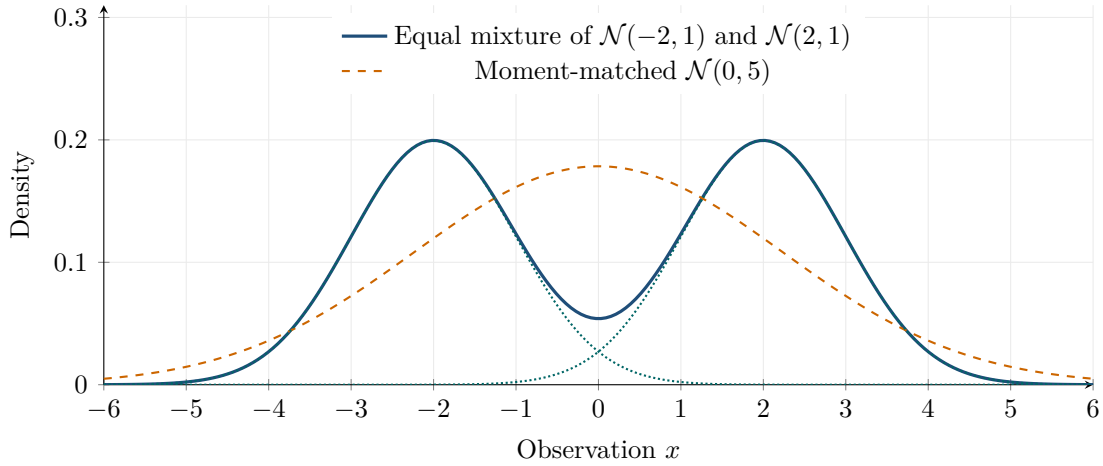


Figure 2.4: Pooling two biased sensors produces the blue mixture density. The dotted curves are the weighted sensor densities, which add to the blue curve. A single Gaussian with the same mean and variance puts substantially more probability near zero.

For example, in sensor 1 the event  $-1 \leq X \leq 1$  corresponds to a standardized error between one and three. Sensor 2 gives the same probability by symmetry. Therefore the pooled probability is

$$\mathbb{P}(|X| \leq 1) = \Phi(3) - \Phi(1) \approx 0.1573.$$

The Gaussian  $\mathcal{N}(0, 5)$  instead gives  $2\Phi(1/\sqrt{5}) - 1 \approx 0.3453$ . Knowing the mean and variance alone does not determine interval probabilities.

Also distinguish selecting one sensor from averaging two readings. If we take one independent reading from *each* sensor and average them, their opposite biases cancel:

$$\frac{X_1 + X_2}{2} \sim \mathcal{N}(0, 1/2).$$

A mixture chooses which source supplies an observation. An average combines observations from both sources. These are different experiments.

### 2.9.4 Different users have different message rates

Two sensors give a small number of groups. In a user population, the differences can vary continuously. Some users receive very few messages; others receive many. This variation between members of a population is called *heterogeneity*.

Observe each user for one day. Let  $\Lambda$  be the user's expected number of messages in that day. A particular user's value, denoted  $\lambda$ , is fixed during the observation; the uppercase variable  $\Lambda$  varies when we choose a different user. Assume that, for a user with value  $\lambda$ ,

$$N \mid \Lambda = \lambda \sim \text{Poisson}(\lambda).$$

Thus even for a fixed user the realized count  $N$  fluctuates from day to day. Across users there is additional variation in the mean count.

The within-user mean and variance both equal  $\Lambda$ . Averaging over users, the laws of total expectation and variance give

$$\mathbb{E}[N] = \mathbb{E}[\Lambda], \quad \text{Var}(N) = \mathbb{E}[\Lambda] + \text{Var}(\Lambda).$$

The first contribution to the variance is daily Poisson variation. The second is variation in the users' average activity. If the latter is positive, the pooled variance exceeds the pooled mean. This is called *overdispersion* relative to a Poisson distribution.

Replacing every user's rate by the population average removes this second contribution. It also misses the extra zeros contributed by quiet users and the large counts contributed by busy users.

### 2.9.5 A Gamma distribution of Poisson means

We need a distribution for the positive quantity  $\Lambda$ . One possible choice is the Gamma family, whose shape and rate let us describe different amounts of variation among users:

$$\Lambda \sim \text{Gamma}(\alpha, \beta).$$

The Gamma moments immediately give

$$\mathbb{E}[N] = \frac{\alpha}{\beta}, \quad \text{Var}(N) = \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2}.$$

For  $\alpha = 2, \beta = 1/2$ , the mean user count is four and the variance between user means is eight. The pooled count therefore has mean four and variance  $4 + 8 = 12$ .

We can also find the full count distribution. Let  $g(\lambda)$  be the Gamma density of user means. To find the probability of  $k$  messages, average the Poisson probability over all possible user means:

$$\mathbb{P}(N = k) = \int_0^\infty \mathbb{P}(N = k \mid \Lambda = \lambda) g(\lambda) d\lambda.$$

This is the continuous version of the weighted sum for the sensors: possible user means replace the discrete source labels, and the integral replaces the sum. The count remains discrete; it is the mixing parameter that varies continuously.

Substituting the Poisson PMF and Gamma density gives

$$\begin{aligned} \mathbb{P}(N = k) &= \int_0^\infty \frac{e^{-\lambda} \lambda^k}{k!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} d\lambda \\ &= \frac{\Gamma(k + \alpha)}{\Gamma(\alpha) k!} \left( \frac{\beta}{\beta + 1} \right)^\alpha \left( \frac{1}{\beta + 1} \right)^k, \quad k = 0, 1, \dots \end{aligned}$$

The second line follows by combining the powers and exponentials, then using the gamma integral with shape  $k + \alpha$  and rate  $\beta + 1$ . The resulting distribution is called *negative binomial*. Here its meaning is supplied by the construction: a Poisson count whose mean differs from user to user according to a Gamma distribution.

For our numerical example,

$$\mathbb{P}(N = 0) = \left( \frac{1/2}{1 + 1/2} \right)^2 = \frac{1}{9} \approx 0.1111.$$

A single  $\text{Poisson}(4)$  distribution instead gives  $e^{-4} \approx 0.0183$ . The mean counts agree, but the predicted fraction of users receiving no messages is very different.

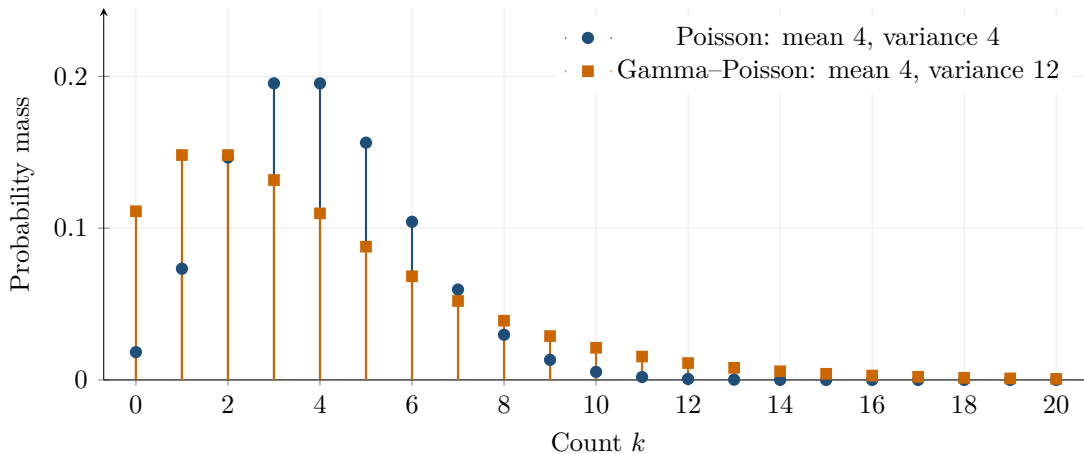


Figure 2.5: Both populations average four messages per user per day. With a common Poisson mean of four, the variance is four. With Gamma-distributed user means, the variance is twelve, with more users receiving zero messages or unusually many messages.

The conditional-moment rules used here extend the finite-mixture identities by replacing weighted sums with integrals. They require finite moments, but they do not require us to evaluate the full mixture PMF before calculating its mean and variance.

### 2.9.6 Ten responses from one user or one response from ten users?

Heterogeneity also matters for binary observations, but the way we group observations changes the result. Suppose half the users click with probability 0.1, and half with probability 0.9. A randomly chosen user's response is a click with probability

$$\frac{1}{2}(0.1) + \frac{1}{2}(0.9) = 0.5.$$

That overall probability does not tell us how much the individual user probabilities differ.

Now compare two experiments. In the first, choose one user and show them ten posts, with independent responses once their click probability is fixed. The chosen user is likely to click about once or about nine times, depending on which group they belong to. In the second, choose ten users independently and record one response from each. Each response is independently a click with probability 0.5, so the count is  $\text{Binomial}(10, 0.5)$ .

The means are both five. The variances are different. Write  $P$  for the first experiment's shared user probability and  $S$  for the ten-response count. Within either user group, the variance is

$10(0.1)(0.9) = 0.9$ . The group means are one and nine, whose variance around five is sixteen. Therefore

$$\text{Var}(S) = 0.9 + 16 = 16.9,$$

whereas the second experiment has variance  $10(0.5)(0.5) = 2.5$ .

More generally, let  $q = \mathbb{E}[P]$  and suppose  $S \mid P \sim \text{Binomial}(n, P)$ . Then

$$\begin{aligned}\mathbb{E}[S] &= nq, \\ \text{Var}(S) &= n\mathbb{E}[P(1-P)] + n^2 \text{Var}(P) \\ &= nq(1-q) + n(n-1) \text{Var}(P).\end{aligned}$$

The final step uses  $\mathbb{E}[P(1-P)] = q(1-q) - \text{Var}(P)$ , obtained by expanding  $\mathbb{E}[P^2] = \text{Var}(P) + q^2$ . The extra variance comes from sharing the same random user probability across trials.

If a fresh  $P_i$  is independently drawn for each trial instead, and the responses are independently generated given those probabilities, the responses are independent Bernoulli variables with parameter  $q$ . Their sum is  $\text{Binomial}(n, q)$ . Which quantities are shared across observations is part of specifying the experiment.

## 2.10 Extension: Gaussian scale mixtures and Student's $t$

Return to measurement errors. So far the two sensors differed in their average errors. Another possibility is that a sensor is usually precise but occasionally operates under much noisier conditions, without changing its average error.

For example, suppose 95% of readings have error  $\mathcal{N}(\mu, \sigma^2)$ , and the remaining 5% have standard deviation  $5\sigma$ :

$$X \sim 0.95 \mathcal{N}(\mu, \sigma^2) + 0.05 \mathcal{N}(\mu, 25\sigma^2).$$

This is called a *contaminated Gaussian* model. The noisier readings make large errors more common. The mean stays  $\mu$ , while the mixture variance is

$$\text{Var}(X) = 0.95\sigma^2 + 0.05(25\sigma^2) = 2.2\sigma^2.$$

Instead of two noise levels, we can allow a continuous range. Describe the noise using its inverse variance  $\tau$ , often called *precision*. A small  $\tau$  means a large variance and a noisy reading; a large  $\tau$  means a narrow error distribution. For a standardized error, use

$$Z \mid \tau \sim \mathcal{N}(0, 1/\tau).$$

The Gamma family is a natural candidate for a positive random precision. Choose

$$\tau \sim \text{Gamma}(\nu/2, \nu/2), \quad \nu > 0.$$

Using equal shape and rate fixes  $\mathbb{E}[\tau] = 1$ ; the Gamma variance formula gives  $\text{Var}(\tau) = 2/\nu$ . Thus  $\nu$  controls how much the noise level varies: large  $\nu$  keeps precision close to one, while small  $\nu$  permits more variation.

Average the conditional Gaussian density over these precision values, just as we averaged Poisson probabilities over user means. Let  $g_\tau$  denote the Gamma density and  $f_{Z|\tau}(z \mid u)$  the

Gaussian density with variance  $1/u$ . Then

$$\begin{aligned} f_Z(z) &= \int_0^\infty f_{Z|\tau}(z | u) g_\tau(u) du \\ &= \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi} \Gamma(\nu/2)} \left(1 + \frac{z^2}{\nu}\right)^{-(\nu+1)/2}. \end{aligned}$$

The product inside the integral is proportional, as a function of  $u$ , to  $u^{(\nu-1)/2} e^{-(\nu+z^2)u/2}$ . This is another gamma integral, now with shape  $(\nu+1)/2$  and rate  $(\nu+z^2)/2$ . The resulting density defines *Student's  $t$ -distribution*, written  $Z \sim t_\nu$ . In this construction,  $\nu$  controls variation in precision and therefore how often extreme errors occur.

Occasionally drawing a small precision produces an unusually large Gaussian error. After averaging over precisions, extreme errors remain much more likely than under a standard Gaussian. This is the meaning of *heavy tails* here: for large  $|z|$ , the density decreases like  $|z|^{-(\nu+1)}$ , rather than like  $e^{-z^2/2}$ .

For  $\nu > 1$ , the mean exists and is zero. For  $\nu > 2$ , total variance and the Gamma integral give

$$\text{Var}(Z) = \mathbb{E}[1/\tau] = \frac{\nu}{\nu-2}.$$

When  $1 < \nu \leq 2$ , rare large observations make the second-moment integral diverge, so the variance is infinite. When  $0 < \nu \leq 1$ , even the absolute first moment diverges and the mean is undefined. This illustrates why the convergence conditions in the definitions of expectation and variance matter.

For an error with center  $\mu$  and scale  $s > 0$ , write  $X = \mu + sZ$ . For  $\nu > 2$ ,

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = s^2 \frac{\nu}{\nu-2}.$$

The scale  $s$  is therefore not its standard deviation. As  $\nu$  increases, precision concentrates near one and the standard Student distribution approaches  $\mathcal{N}(0, 1)$ .

## 2.11 Distribution reference

The table collects the means and variances derived above. The support lists the values the variable can take, distinguishing counts from waiting times and signed errors.

| Distribution                  | Support       | $\mathbb{E}[X]$ | $\text{Var}(X)$           |
|-------------------------------|---------------|-----------------|---------------------------|
| $\text{Exp}(\lambda)$         | $[0, \infty)$ | $1/\lambda$     | $1/\lambda^2$             |
| $\text{Poisson}(m)$           | $0, 1, \dots$ | $m$             | $m$                       |
| $\text{Gamma}(\alpha, \beta)$ | $[0, \infty)$ | $\alpha/\beta$  | $\alpha/\beta^2$          |
| $\text{Bernoulli}(p)$         | $0, 1$        | $p$             | $p(1-p)$                  |
| $\text{Binomial}(n, p)$       | $0, \dots, n$ | $np$            | $np(1-p)$                 |
| $\text{Geometric}(p)$         | $1, 2, \dots$ | $1/p$           | $(1-p)/p^2$               |
| $\mathcal{N}(\mu, \sigma^2)$  | $\mathbb{R}$  | $\mu$           | $\sigma^2$                |
| $t_\nu$                       | $\mathbb{R}$  | $0 \ (\nu > 1)$ | $\nu/(\nu-2) \ (\nu > 2)$ |

Here  $\lambda, \alpha, \beta, m, \nu > 0$ ,  $\sigma > 0$ , and  $n$  is a nonnegative integer. The Bernoulli and Binomial parameters allow  $0 \leq p \leq 1$ ; the Geometric requires  $0 < p \leq 1$ . Gamma uses shape and rate. Geometric counts trials including the first success. Student moments exist only under the conditions shown.

| Connection                   | Required mechanism or condition                                                                |
|------------------------------|------------------------------------------------------------------------------------------------|
| Exponential to Gamma         | Sum independent waiting times with a common rate.                                              |
| Exponential to Poisson       | Count arrivals generated by independent exponential interarrival times.                        |
| Bernoulli to Binomial        | Sum a fixed number of independent trials with the same success probability.                    |
| Binomial to Poisson          | Take $n \rightarrow \infty$ , $p \rightarrow 0$ , with $np = m$ .                              |
| Geometric to Exponential     | Rescale trial duration to $\Delta$ , set $p = \lambda\Delta$ , and let $\Delta \downarrow 0$ . |
| Poisson to negative binomial | Mix the Poisson rate with a Gamma distribution.                                                |
| Gaussian to Student          | Mix the Gaussian precision with $\text{Gamma}(\nu/2, \nu/2)$ .                                 |

## Chapter 3

# Joint Distributions, Dependence, and Random Vectors

## 3.1 Two observations from the same user

Show two posts to one randomly chosen user. Let  $X$  indicate a click on the first post and  $Y$  a click on the second. Each variable is either zero or one. Knowing the click probability for each post does not yet tell us the probability of two clicks: we also need to know how the responses are related.

Use the population from Section 2.9: half the users have click probability 0.1, and half have click probability 0.9. Once a user is selected, their two responses are independent with that user's probability. Write  $P$  for this shared probability.

For either post, the marginal click probability is

$$\mathbb{P}(X = 1) = \mathbb{P}(Y = 1) = \frac{1}{2}(0.1) + \frac{1}{2}(0.9) = 0.5.$$

But the probability of two clicks is

$$\mathbb{P}(X = 1, Y = 1) = \frac{1}{2}(0.1)^2 + \frac{1}{2}(0.9)^2 = 0.41,$$

not  $0.5 \times 0.5 = 0.25$ . The same user supplies both responses, so both share the same tendency to click. Similarly,

$$\mathbb{P}(X = 1, Y = 0) = \frac{1}{2}(0.1)(0.9) + \frac{1}{2}(0.9)(0.1) = 0.09.$$

The four possible pairs give the following table.

|              | $Y = 0$ | $Y = 1$ | Row total |
|--------------|---------|---------|-----------|
| $X = 0$      | 0.41    | 0.09    | 0.50      |
| $X = 1$      | 0.09    | 0.41    | 0.50      |
| Column total | 0.50    | 0.50    | 1         |

### 3.1.1 Joint and marginal probabilities

The entries inside the table form the *joint probability mass function*:

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

The comma means that both equalities hold in the same repetition of the experiment. A joint PMF must be nonnegative and satisfy  $\sum_x \sum_y p_{X,Y}(x, y) = 1$ . To find the probability of an event, add the cells that satisfy it. For example,

$$\mathbb{P}(X = Y) = 0.41 + 0.41 = 0.82.$$

If we only care about the first response, either value of the second is acceptable. We therefore add across a row:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y). \quad (3.1)$$

These are the *marginal* PMFs, written in the margins of the table. Marginalizing means ignoring a variable by summing over its possible values, not setting it to zero.

If the two responses instead came from independently selected users, every interior cell would be 0.25. The row and column totals would still be 0.5. Thus the same two marginal distributions can belong to very different joint distributions.

### 3.1.2 Conditioning selects and renormalizes a row

Suppose we observe a click on the first post. Among the cases in the  $X = 1$  row, the fraction with a second click is

$$\mathbb{P}(Y = 1 \mid X = 1) = \frac{0.41}{0.50} = 0.82.$$

After a first non-click, it is instead

$$\mathbb{P}(Y = 1 \mid X = 0) = \frac{0.09}{0.50} = 0.18.$$

Each conditional distribution divides the selected row by its total so that its entries add to one. In general, for  $p_X(x) > 0$ ,

$$p_{Y|X}(y \mid x) = \frac{p_{X,Y}(x, y)}{p_X(x)}, \quad p_{X,Y}(x, y) = p_X(x)p_{Y|X}(y \mid x). \quad (3.2)$$

The first click is informative because it changes which type of user we think we selected. Bayes' theorem gives

$$\mathbb{P}(P = 0.9 \mid X = 1) = \frac{0.9(0.5)}{0.5} = 0.9.$$

Averaging the second-click probability over this updated population gives  $0.9(0.9) + 0.1(0.1) = 0.82$ , as the table did. Nothing in this model says that the first click *causes* the second: both reflect the same unobserved user probability.

### 3.1.3 Independence means the joint distribution factors

For independent responses, learning  $X$  would leave the distribution of  $Y$  unchanged. In a discrete model this is equivalent to

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y. \quad (3.3)$$

For general random variables, independence means

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

for all measurable sets  $A, B$ . The PMF condition checks this using individual cells. We write  $X \perp\!\!\!\perp Y$ .

In our example, the responses are *conditionally independent* given  $P$ : for either fixed user probability, their conditional joint PMF factors. They are not independent when that probability is unknown and averaged out. This distinction explains why repeated observations from one person can behave differently from observations from different people.

## 3.2 Joint densities: probability over a region

For a pair with a joint density, replace the probability table by a density over the plane. Near  $(x, y)$ , a small rectangle of side lengths  $\Delta x, \Delta y$  has probability approximately  $f_{X,Y}(x, y)\Delta x\Delta y$ . Adding small rectangles becomes a double integral. For a region  $R$ ,

$$\mathbb{P}((X, Y) \in R) = \iint_R f_{X,Y}(x, y) dx dy.$$

The density is nonnegative, and its integral over the whole plane is one.

### 3.2.1 Two arrival times, put in order

Suppose two people each independently choose an arrival time uniformly within a one-hour window. Let  $U, V$  be their times in hours after the window begins. Their unordered pair is uniform on the unit square:  $f_{U,V}(u, v) = 1$  for  $0 < u, v < 1$ .

Now record the earlier and later times,

$$T_1 = \min(U, V), \quad T_2 = \max(U, V).$$

For  $t_1 < t_2$ , the ordered pair near  $(t_1, t_2)$  can arise from  $(U, V)$  near either  $(t_1, t_2)$  or  $(t_2, t_1)$ . These two equally sized regions contribute equally. Hence

$$f_{T_1, T_2}(t_1, t_2) = \begin{cases} 2, & 0 < t_1 < t_2 < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The allowed triangle has area  $1/2$ ; multiplying by the density two gives total probability one. Sorting has introduced dependence: the later time cannot be smaller than the earlier time.

For instance, both arrivals occur in the first half-hour exactly when  $T_2 \leq 1/2$ . The relevant triangle has area  $\frac{1}{2}(\frac{1}{2})^2 = 1/8$ , so

$$\mathbb{P}(T_2 \leq 1/2) = 2 \cdot \frac{1}{8} = \frac{1}{4}.$$

This agrees with the probability that both independent original times fall in the first half of the window.

### 3.2.2 Marginalizing integrates along a slice

Fix an earlier time  $t$ . The later time can range from  $t$  to one, so the marginal density of the earlier time is

$$f_{T_1}(t) = \int_t^1 2 ds = 2(1 - t), \quad 0 < t < 1.$$

There are more possible later arrivals when the first arrival is early. Conversely, if the later arrival is at  $s$ , the earlier one can be anywhere between zero and  $s$ :

$$f_{T_2}(s) = \int_0^s 2 dt = 2s, \quad 0 < s < 1.$$

The general marginalization rules are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx. \quad (3.4)$$

Although the bounds are written over the whole line, the integrand is zero outside the joint support. The triangular support is what produced the variable integration limits in the example.

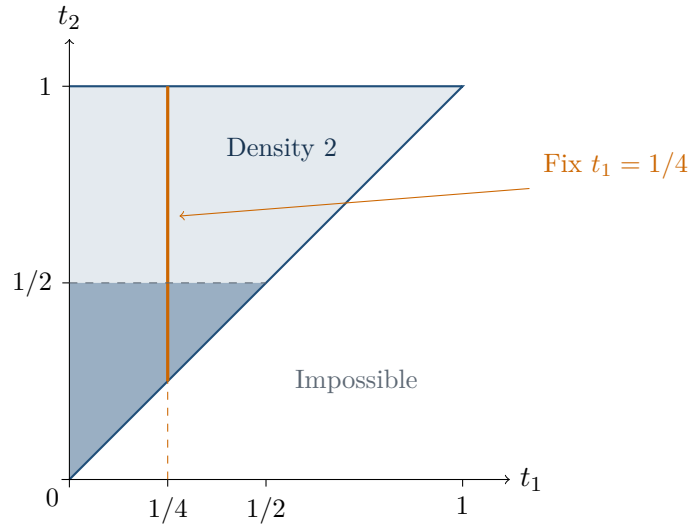


Figure 3.1: Every possible ordered pair lies above the diagonal  $t_2 = t_1$ . The dark triangle represents two arrivals before half an hour. At a fixed  $t_1 = 1/4$ , the later time ranges along the marked vertical segment from  $1/4$  to one. Marginalizing adds density along such a segment; conditioning normalizes the segment's density.

### 3.2.3 A conditional density is a normalized slice

If the first arrival is at time  $t$ , how much longer do we wait for the second? Dividing the joint density by the earlier-time marginal gives

$$f_{T_2|T_1}(s | t) = \frac{2}{2(1-t)} = \frac{1}{1-t}, \quad t < s < 1.$$

Thus, conditional on the earlier time, the later time is uniform over the remaining part of the window. At  $t = 1/4$ , for example,

$$\mathbb{P}(T_2 > 3/4 | T_1 = 1/4) = \int_{3/4}^1 \frac{1}{1-1/4} ds = \frac{1}{3}.$$

In general, wherever  $f_X(x) > 0$ , the conditional-density formula is

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}. \quad (3.5)$$

This is not a division by  $\mathbb{P}(X = x)$ , which is zero for a continuous variable. One way to understand the formula is to condition on a narrow interval around  $x$ , then shrink that interval. In a smooth density model this leaves the normalized slice above. Conditional densities need only be specified for almost every conditioning value, meaning that changing them on a set of  $X$ -probability zero has no effect.

For a pair with a joint density, independence is equivalent to  $f_{X,Y}(x, y) = f_X(x)f_Y(y)$  almost everywhere. The ordered times fail this test: both marginals are positive inside  $(0, 1)$ , but their joint density is zero whenever  $t_1 > t_2$ .

### 3.3 Averaging a function of two observations

Sometimes we care about a quantity computed from the pair, such as the total clicks  $X + Y$ , the indicator of two clicks  $XY$ , or the gap  $T_2 - T_1$ . As in Section 2.2, calculate the quantity at each outcome and average it with the outcome's probability:

$$\mathbb{E}[g(X, Y)] = \begin{cases} \sum_x \sum_y g(x, y) p_{X,Y}(x, y), & \text{discrete pair,} \\ \iint_{\mathbb{R}^2} g(x, y) f_{X,Y}(x, y) dx dy, & \text{pair with a density.} \end{cases} \quad (3.6)$$

We require  $\mathbb{E}[|g(X, Y)|] < \infty$  for a finite expectation. There is no need to derive the distribution of  $g(X, Y)$  first.

For the click pair,  $XY$  is one only in the  $(1, 1)$  cell, so

$$\mathbb{E}[XY] = 0.41, \quad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] = 1.$$

Linearity applies to the sum even though the clicks are dependent. For the ordered arrival times, the average gap is

$$\begin{aligned} \mathbb{E}[T_2 - T_1] &= \int_0^1 \int_t^1 (s - t) 2 ds dt \\ &= \int_0^1 (1 - t)^2 dt = \frac{1}{3} \text{ hour.} \end{aligned}$$

The separate means give the same answer:  $\mathbb{E}[T_1] = 1/3$  and  $\mathbb{E}[T_2] = 2/3$ , obtained by integrating against their marginal densities.

If  $X, Y$  are independent and have finite absolute first moments, their joint probabilities factor, so

$$\begin{aligned} \mathbb{E}[XY] &= \sum_x \sum_y xy p_X(x) p_Y(y) \\ &= \left( \sum_x x p_X(x) \right) \left( \sum_y y p_Y(y) \right) = \mathbb{E}[X] \mathbb{E}[Y]. \end{aligned}$$

Integrals give the same result for independent variables with densities. This factorization also applies to integrable functions of independent variables:  $\mathbb{E}[a(X)b(Y)] = \mathbb{E}[a(X)]\mathbb{E}[b(Y)]$ . The click table shows why it cannot be assumed without independence.

### 3.4 Covariance and correlation

In Chapter 2, covariance appeared as the extra term in the variance of a sum. We can now compute it from a joint distribution and examine what it does, and does not, tell us.

### 3.4.1 Do deviations reinforce or cancel each other?

Let  $\mu_X = \mathbb{E}[X]$  and  $\mu_Y = \mathbb{E}[Y]$ , and assume both variables have finite second moments. When both observations are above their means, or both are below, the product  $(X - \mu_X)(Y - \mu_Y)$  is positive. When they deviate in opposite directions, it is negative. Covariance averages these products:

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].\end{aligned}\tag{3.7}$$

The second line follows by expanding the product and taking expectations.

For the two clicks,

$$\text{Cov}(X, Y) = 0.41 - (0.5)(0.5) = 0.16.$$

The agreement cells  $(0, 0)$  and  $(1, 1)$  have more probability than the disagreement cells. Their positive centered products dominate. Consequently, the two-click total is more variable than the total of two independent Bernoulli(0.5) variables:

$$\text{Var}(X + Y) = 0.25 + 0.25 + 2(0.16) = 0.82,$$

instead of 0.5. Indeed, the total is zero or two with probability 0.41 each, and one with probability 0.18.

Covariance depends on units. If  $X$  is measured in minutes and  $Y$  in bytes, covariance is measured in minute-bytes. More generally,

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y).$$

Shifts do not change deviations; scaling changes their product.

### 3.4.2 Correlation removes the units

For positive standard deviations  $\sigma_X, \sigma_Y$ , define

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \mathbb{E}\left[\frac{X - \mu_X}{\sigma_X} \frac{Y - \mu_Y}{\sigma_Y}\right].\tag{3.8}$$

This is the average product of the standardized deviations. Changing minutes to seconds no longer affects the answer. Reversing the sign of one variable reverses the sign of the correlation. For our clicks,  $\sigma_X = \sigma_Y = 0.5$ , hence  $\rho_{X,Y} = 0.64$ .

To see why correlation lies between  $-1$  and one, put  $U = (X - \mu_X)/\sigma_X$  and  $V = (Y - \mu_Y)/\sigma_Y$ . Both have mean zero and variance one, so

$$0 \leq \mathbb{E}[(U - V)^2] = 2 - 2\rho_{X,Y}, \quad 0 \leq \mathbb{E}[(U + V)^2] = 2 + 2\rho_{X,Y}.$$

At  $\rho = 1$ , the first square has expectation zero, so  $U = V$  with probability one. At  $\rho = -1$ ,  $U = -V$  with probability one. Thus perfect correlation means an exact affine relationship, with positive or negative slope. Correlation is undefined if either variable has variance zero: a constant has no fluctuations to standardize.

### 3.4.3 Zero covariance can hide complete dependence

Independence implies zero covariance because expectations of products factor. The reverse implication fails: positive and negative contributions to covariance can cancel even when one variable determines the other.

Let  $X$  be uniform on  $[-1, 1]$ , and let  $Y = X^2$ . Then

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[Y] = \mathbb{E}[X^2] = \frac{1}{3}, \quad \mathbb{E}[XY] = \mathbb{E}[X^3] = 0.$$

The odd powers average to zero by symmetry. Thus  $\text{Cov}(X, Y) = 0$ , and the correlation is zero since both variances are positive. Yet knowing  $X$  tells us  $Y$  exactly.

An event calculation also rules out independence. Here  $Y \leq 1/4$  is exactly the event  $|X| \leq 1/2$ , so

$$\mathbb{P}(Y \leq 1/4 \mid |X| \leq 1/2) = 1, \quad \mathbb{P}(Y \leq 1/4) = \frac{1}{2}.$$

Correlation detects alignment of signed deviations, not every possible relationship. An upward-then-downward or U-shaped relationship can have zero correlation while remaining highly informative.

## 3.5 Conditional expectation and two sources of variation

### 3.5.1 A mean that depends on what we observe

Before seeing the first response, our average prediction for the second click is 0.5. After seeing it, the relevant average is 0.18 or 0.82. The function

$$m(x) = \mathbb{E}[Y \mid X = x]$$

assigns a mean to each conditional distribution. In the click example,

$$m(0) = 0.18, \quad m(1) = 0.82.$$

For fixed  $x$ ,  $m(x)$  is a number. Before  $X$  is observed,  $m(X)$  is a random variable. This random variable is denoted  $\mathbb{E}[Y \mid X]$ . The same distinction applies to  $\mathbb{E}[X \mid Y]$ : the variable after the bar specifies the information available, and the variable before it is averaged.

The conditional mean uses the usual expectation formula with a conditional distribution:

$$m(x) = \begin{cases} \sum y p_{Y|X}(y \mid x), & \text{discrete,} \\ \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x) dy, & \text{with a conditional density.} \end{cases}$$

For a binary outcome, its conditional mean is simply its conditional probability of being one.

For the ordered arrival times,  $T_2 \mid T_1 = t$  is uniform on  $(t, 1)$ , so its mean is the midpoint:

$$\mathbb{E}[T_2 \mid T_1 = t] = \frac{t+1}{2}, \quad \mathbb{E}[T_2 - T_1 \mid T_1 = t] = \frac{1-t}{2}.$$

If the first person arrives at minute fifteen, the expected remaining wait is  $(1 - 1/4)/2 = 3/8$  hour, or 22.5 minutes.

### 3.5.2 Averaging conditional means returns the overall mean

Half of first responses are clicks. Averaging the two corresponding second-click probabilities gives

$$\mathbb{E}[m(X)] = \frac{1}{2}(0.18) + \frac{1}{2}(0.82) = 0.5 = \mathbb{E}[Y].$$

This is the law of total expectation from Chapter 2, now applied to an observed response rather than a hidden group label. For integrable  $Y$ ,

$$\mathbb{E}[\mathbb{E}[Y | X]] = \mathbb{E}[Y]. \quad (3.9)$$

In a discrete model the reason is visible by expanding both averages:

$$\sum_x \left( \sum_y y p_{Y|X}(y | x) \right) p_X(x) = \sum_x \sum_y y p_{X,Y}(x, y) = \mathbb{E}[Y].$$

For a pair with a density, replace the sums by integrals. For the arrival example,

$$\mathbb{E}[T_2] = \mathbb{E}\left[\frac{T_1 + 1}{2}\right] = \frac{1/3 + 1}{2} = \frac{2}{3}.$$

There is no independence assumption: the formula explicitly accounts for how the conditional distribution changes with  $X$ .

### 3.5.3 Within-group variation and variation of group means

The conditional variance is

$$v(x) = \text{Var}(Y | X = x) = \mathbb{E}[(Y - m(x))^2 | X = x].$$

It measures the uncertainty left after the value of  $X$  is known. The notation  $\text{Var}(Y | X) = v(X)$  is again a random variable.

Write  $\mu = \mathbb{E}[Y]$ . Each deviation from the overall mean splits as

$$Y - \mu = \underbrace{Y - m(X)}_{\text{deviation from conditional mean}} + \underbrace{m(X) - \mu}_{\text{shift of conditional mean}}.$$

Within a group with fixed  $X = x$ , the first term has mean zero, and the second term is constant. Their product therefore has conditional mean zero. Square the decomposition and average, first within groups and then over  $X$ , to obtain

$$\boxed{\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X])}. \quad (3.10)$$

This holds when  $Y$  has a finite second moment. It generalizes the within-source and between-source decomposition in (2.7) to arbitrary conditioning information.

In the click example, both conditional Bernoulli variances are  $0.18(0.82) = 0.1476$ . The conditional means lie 0.32 on either side of 0.5, giving

$$\underbrace{0.25}_{\text{Var}(Y)} = \underbrace{0.1476}_{\mathbb{E}[\text{Var}(Y|X)]} + \underbrace{0.1024}_{\text{Var}(\mathbb{E}[Y|X])}.$$

Observing the first click removes some uncertainty about the second, but it does not determine the second response.

Total variance implies that the *average* conditional variance is at most the unconditional variance. It does not require every group to have smaller variance. In the noisy-sensor mixture of Section 2.10, identifying a reading as coming from the noisy component gives variance  $25\sigma^2$ , larger than the pooled variance  $2.2\sigma^2$ .

### 3.5.4 Why the conditional mean is useful for prediction

Suppose we predict  $Y$  using only  $X$ , with a rule  $a(X)$ . Inside a fixed group  $X = x$ , expanding a square gives

$$\mathbb{E}[(Y - a(x))^2 \mid X = x] = v(x) + (m(x) - a(x))^2.$$

The cross term disappears because the deviations from  $m(x)$  have conditional mean zero. Consequently, for a rule with finite mean squared error,

$$\mathbb{E}[(Y - a(X))^2] = \mathbb{E}[v(X)] + \mathbb{E}[(m(X) - a(X))^2].$$

The first term is uncertainty left in  $Y$  after observing  $X$ . The second is the extra error from predicting something other than the conditional mean. It is nonnegative and vanishes for  $a(X) = m(X)$ . Thus the conditional mean gives the smallest expected squared prediction error among rules using the same information. For the clicks, the constant rule 0.5 has error 0.25, whereas the conditional-mean rule has error 0.1476. This is a property of the probability model; estimating such a rule from a dataset is a separate statistical problem.

## 3.6 Random vectors and covariance matrices

A pair is the simplest example of several measurements made together. For a single object we might record height, width, temperature, and two sensor readings. Collect  $d$  measurements into a column vector

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_d \end{pmatrix} \in \mathbb{R}^d.$$

This is a *random vector*: all coordinates refer to the same randomly selected object or experimental repetition. A lowercase  $\mathbf{x}$  denotes one realized vector. The joint distribution specifies probabilities for regions in  $\mathbb{R}^d$ .

The mean vector averages each coordinate:

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{pmatrix}.$$

To record how every pair of coordinates fluctuates together, arrange their covariances into a matrix.

**Definition 3.1** (Covariance matrix). For a random vector with finite second moments,

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top], \quad \Sigma_{ij} = \text{Cov}(X_i, X_j). \quad (3.11)$$

The superscript  $\top$  denotes transpose: it turns the column of deviations into a row. Their product is a  $d \times d$  matrix whose  $(i, j)$  entry is the product of the two deviations.

The diagonal entries are variances; the off-diagonal entries are covariances. For the two-click vector,

$$\boldsymbol{\mu} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.25 & 0.16 \\ 0.16 & 0.25 \end{pmatrix}.$$

The matrix is symmetric because  $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$ . If the coordinates have positive variances, standardizing each one gives the correlation matrix, with entries  $R_{ij} = \Sigma_{ij} / \sqrt{\Sigma_{ii}\Sigma_{jj}}$  and ones on its diagonal. For the clicks its off-diagonal entries are 0.64.

### 3.6.1 A covariance matrix cannot imply a negative variance

Take a weighted sum  $S = a_1X_1 + \cdots + a_dX_d = \mathbf{a}^\top \mathbf{X}$ . Expanding its squared centered value gives

$$\text{Var}(S) = \sum_{i=1}^d \sum_{j=1}^d a_i a_j \text{Cov}(X_i, X_j) = \mathbf{a}^\top \Sigma \mathbf{a} \geq 0. \quad (3.12)$$

A symmetric matrix satisfying this inequality for every real vector  $\mathbf{a}$  is called *positive semidefinite*. Every covariance matrix must have this property. It is *positive definite* if the inequality is strict for every nonzero  $\mathbf{a}$ , meaning that no nontrivial weighted combination has zero variance.

For example, the symmetric matrix

$$M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

cannot be a covariance matrix. It would imply  $\text{Var}(X_1 - X_2) = 1 + 1 - 2(2) = -2$ . Positive diagonal entries alone do not make a covariance matrix valid. In two dimensions, a symmetric matrix

$$\begin{pmatrix} v_1 & c \\ c & v_2 \end{pmatrix}$$

is positive semidefinite exactly when  $v_1 \geq 0$ ,  $v_2 \geq 0$ , and  $c^2 \leq v_1 v_2$ . For positive variances, the last condition is the correlation bound  $|\rho| \leq 1$ .

A diagonal covariance matrix means that distinct coordinates are uncorrelated. It does not generally mean that they are independent:  $(X, X^2)$ , with  $X$  uniform on  $[-1, 1]$ , remains a counterexample.

## 3.7 Linear transformations of random vectors

Two sensors measuring the same quantity can be summarized by their average and their difference. Converting units, subtracting a known bias, or combining features into a score are other examples of linear combinations. We want rules for the means and covariances of the results.

Let  $A$  be a fixed  $k \times d$  matrix and  $\mathbf{b} \in \mathbb{R}^k$  a fixed vector, and set

$$\mathbf{W} = A\mathbf{X} + \mathbf{b}.$$

Row  $i$  of  $A$  supplies the weights for the  $i$ -th new variable. Linearity of expectation gives

$$\mathbb{E}[\mathbf{W}] = A\boldsymbol{\mu} + \mathbf{b}. \quad (3.13)$$

Subtracting this mean yields  $\mathbf{W} - \mathbb{E}[\mathbf{W}] = A(\mathbf{X} - \boldsymbol{\mu})$ . Hence

$$\begin{aligned} \text{Cov}(\mathbf{W}) &= \mathbb{E}[A(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top A^\top] \\ &= A\Sigma A^\top. \end{aligned} \quad (3.14)$$

Here  $\text{Cov}(\mathbf{W})$  denotes the covariance matrix of a vector. These rules need finite second moments, not independence or Gaussian distributions.

**Example 3.2** (Average error and disagreement between sensors). Let  $X_1, X_2$  be centered sensor errors with variances one and covariance 0.8. Define

$$M = \frac{X_1 + X_2}{2}, \quad D = X_1 - X_2, \quad A = \begin{pmatrix} 1/2 & 1/2 \\ 1 & -1 \end{pmatrix}.$$

Both new means are zero, and matrix multiplication gives

$$\text{Cov} \begin{pmatrix} M \\ D \end{pmatrix} = A \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix} A^\top = \begin{pmatrix} 0.9 & 0 \\ 0 & 0.4 \end{pmatrix}.$$

In scalar form,

$$\text{Var}(M) = \frac{1 + 1 + 2(0.8)}{4} = 0.9, \quad \text{Var}(D) = 1 + 1 - 2(0.8) = 0.4.$$

Positively correlated errors often move together, so averaging them removes little of the shared error. Subtracting them cancels that shared variation. With independent unit-variance errors, the two variances would instead be 0.5 and 2.

The zero covariance between  $M$  and  $D$  follows from the equal original variances. It does not by itself establish independence. The Gaussian model below supplies an important case where it does.

## 3.8 The multivariate Gaussian

### 3.8.1 Construct correlated readings from independent noise

Let  $Z_1, Z_2$  be independent standard Gaussian variables. Consider the sensor-error model

$$X_1 = Z_1, \quad X_2 = 0.8Z_1 + 0.6Z_2.$$

Both sensors contain the disturbance  $Z_1$ ; the second also has an independent disturbance  $Z_2$ . The known rules for independent Gaussian sums give

$$\mathbb{E}[X_1] = \mathbb{E}[X_2] = 0, \quad \text{Var}(X_1) = 1, \quad \text{Var}(X_2) = 0.8^2 + 0.6^2 = 1,$$

and

$$\text{Cov}(X_1, X_2) = 0.8 \text{Var}(Z_1) + 0.6 \text{Cov}(Z_1, Z_2) = 0.8.$$

Thus both marginal distributions are standard Gaussian, but their joint distribution is correlated. It is a *bivariate Gaussian*.

The construction extends to any number of measurements.

**Definition 3.3** (Multivariate Gaussian). Let  $\mathbf{Z}$  have  $m$  independent standard Gaussian coordinates, and let  $A$  be a fixed  $d \times m$  matrix. A vector of the form

$$\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z}$$

has a multivariate Gaussian distribution, written

$$\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma), \quad \Sigma = AA^\top.$$

Its mean vector is  $\boldsymbol{\mu}$ , and its covariance matrix is  $\Sigma$ .

The independent noises supply the randomness; the rows of  $A$  specify which noises each measurement shares. Equivalently, a vector is jointly Gaussian if every real linear combination of its coordinates is a univariate Gaussian, allowing a constant when the variance is zero. For the construction above this follows because each combination is itself a sum of independent Gaussian noises. In particular,

$$\mathbf{a}^\top \mathbf{X} \sim \mathcal{N}(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \Sigma \mathbf{a}). \quad (3.15)$$

### 3.8.2 From circular to elliptical density contours

Independence gives the joint density of two standard noises:

$$f_{Z_1, Z_2}(z_1, z_2) = \phi(z_1)\phi(z_2) = \frac{1}{2\pi} \exp\left[-\frac{z_1^2 + z_2^2}{2}\right].$$

It has the same value wherever  $z_1^2 + z_2^2$  is constant, so its equal-density contours are circles centered at zero. These contours describe density values, not a claim that all observations lie inside a particular circle.

For a correlation  $-1 < \rho < 1$ , replace the numerical sensor model by

$$X_1 = Z_1, \quad X_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2.$$

The inverse relations are  $z_1 = x_1$  and  $z_2 = (x_2 - \rho x_1)/\sqrt{1 - \rho^2}$ . The transformation scales areas by  $\sqrt{1 - \rho^2}$ , so preserving probabilities requires dividing the density by this factor. Substitution then gives

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{2\pi\sqrt{1 - \rho^2}} \exp\left[-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1 - \rho^2)}\right]. \quad (3.16)$$

The cross term changes which pairs are likely. With positive  $\rho$ , equal-sign deviations are favored; with negative  $\rho$ , opposite-sign deviations are favored. Both marginals are still  $\mathcal{N}(0, 1)$ .

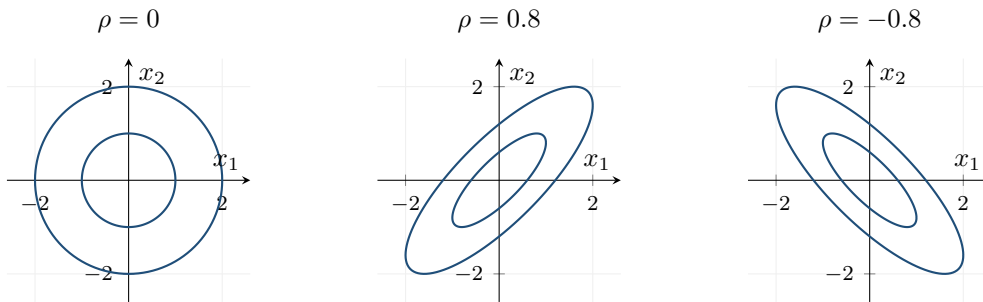


Figure 3.2: Equal-density contours for three bivariate Gaussian models. In every panel both individual features have mean zero and variance one. Correlation changes the joint shape, which separate histograms of the features cannot reveal. Each curve joins points with the same density; the outer curve has lower density than the inner one.

The ellipse directions can be understood without a matrix decomposition. For  $\rho = 0.8$ , use the rotated coordinates

$$U = \frac{X_1 + X_2}{\sqrt{2}}, \quad V = \frac{X_1 - X_2}{\sqrt{2}}.$$

Their variances are  $1 + \rho = 1.8$  and  $1 - \rho = 0.2$ , respectively, and their covariance is zero. In these coordinates the expression in the exponent becomes

$$q = \frac{u^2}{1.8} + \frac{v^2}{0.2}.$$

For fixed  $q = c > 0$ , the ellipse extends by  $\sqrt{1.8c}$  along the equal-sign direction and  $\sqrt{0.2c}$  along the opposite-sign direction. The first is three times longer than the second. Shared sensor fluctuations stretch the distribution along the diagonal.

### 3.8.3 The density in matrix notation

For a positive-definite covariance matrix, the general Gaussian density is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} \sqrt{\det \Sigma}} \exp \left[ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]. \quad (3.17)$$

Here  $\Sigma^{-1}$  is the inverse matrix, and  $\det \Sigma$  is its determinant. These are the matrix versions of dividing by variance and correcting the density for a change of scale. For an invertible square  $A$ , the transformation  $\mathbf{x} = \boldsymbol{\mu} + A\mathbf{z}$  scales volumes by  $|\det A| = \sqrt{\det \Sigma}$ , while

$$\sum_i z_i^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}).$$

This explains both parts of the density formula from the independent standard-noise density.

In two dimensions,

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \quad \det \Sigma = \sigma_1^2 \sigma_2^2 (1 - \rho^2).$$

For positive  $\sigma_1, \sigma_2$  and  $|\rho| < 1$ , putting  $u = (x_1 - \mu_1)/\sigma_1$ ,  $v = (x_2 - \mu_2)/\sigma_2$  gives

$$(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = \frac{u^2 - 2\rho uv + v^2}{1 - \rho^2}.$$

For a diagonal covariance matrix this reduces to a sum of squared standardized deviations, and the density factors into the univariate Gaussian densities.

The quadratic expression is called the *squared Mahalanobis distance*. It measures how unusual a displacement is relative to the model's spread and dependence. In the unit-variance model with  $\rho = 0.8$ , the points  $(1, 1)$  and  $(1, -1)$  have the same ordinary distance from the origin, but

$$q(1, 1) = \frac{2 - 1.6}{0.36} = \frac{10}{9}, \quad q(1, -1) = \frac{2 + 1.6}{0.36} = 10.$$

Disagreement between the sensors is much less typical than a shared positive error. Euclidean distance alone misses that distinction.

### 3.8.4 Marginals, transformations, and the independence exception

Selecting some coordinates is a linear transformation. Their marginal distribution is Gaussian with the corresponding entries of the mean vector and the corresponding rows and columns of the covariance matrix. In particular,  $X_i \sim \mathcal{N}(\mu_i, \Sigma_{ii})$ .

More generally, for fixed  $B$  and  $\mathbf{b}$ ,

$$B\mathbf{X} + \mathbf{b} \sim \mathcal{N}(B\boldsymbol{\mu} + \mathbf{b}, B\Sigma B^\top). \quad (3.18)$$

The result is Gaussian because substituting  $\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z}$  gives another linear transformation of independent standard Gaussian noises. The moment rules in Section 3.7 now describe the *whole distribution*, not just two summaries of it.

For jointly Gaussian variables, zero covariance *does* imply independence. In the nondegenerate case, a diagonal covariance matrix makes the joint density a product of marginal densities. Therefore the average and difference in the sensor example are independent under the joint Gaussian assumption, as well as uncorrelated.

It is not enough for each marginal to be Gaussian. To see why, take  $X \sim \mathcal{N}(0, 1)$  and an independent random sign  $S$ , equal to  $-1$  or  $1$  with probability  $1/2$ . Set  $Y = SX$ . Symmetry makes  $Y \sim \mathcal{N}(0, 1)$ , and

$$\text{Cov}(X, Y) = \mathbb{E}[SX^2] = \mathbb{E}[S]\mathbb{E}[X^2] = 0.$$

Yet  $|Y| = |X|$  always, so they are dependent. The pair is not jointly Gaussian:  $X + Y$  is exactly zero with probability  $1/2$ , but is not a constant. A nondegenerate Gaussian has no such point mass.

Finally, a Gaussian covariance matrix can be singular. If  $X_2 = X_1 \sim \mathcal{N}(0, 1)$ , then

$$\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

All pairs lie on a line. This is a valid Gaussian random vector, but it has no density with respect to two-dimensional area. Formula (3.17) requires a positive-definite matrix; it cannot be used when the determinant is zero.

### 3.9 What one Gaussian measurement tells us about another

Return to  $X_1 = Z_1$ ,  $X_2 = 0.8Z_1 + 0.6Z_2$ . Before seeing the first sensor, the second error has variance one. After observing  $X_1 = x$ , the shared noise  $Z_1$  is known and only the independent noise  $Z_2$  remains:

$$X_2 \mid X_1 = x \sim \mathcal{N}(0.8x, 0.36).$$

If the first sensor reads two units too high, the conditional mean error of the second is 1.6, with standard deviation 0.6. Marginalizing over the first sensor and conditioning on its reading are different operations: the marginal distribution is still  $\mathcal{N}(0, 1)$ .

For the general bivariate Gaussian with positive marginal variances and  $|\rho| < 1$ , write

$$X = \mu_X + \sigma_X Z_1, \quad Y = \mu_Y + \sigma_Y(\rho Z_1 + \sqrt{1 - \rho^2} Z_2).$$

Observing  $X = x$  fixes  $Z_1 = (x - \mu_X)/\sigma_X$ . Therefore

$$Y \mid X = x \sim \mathcal{N}\left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2)\right). \quad (3.19)$$

Equivalently, the conditional mean's slope is  $\text{Cov}(X, Y)/\text{Var}(X)$ . Positive correlation shifts the conditional mean in the direction of the observation; negative correlation shifts it in the

opposite direction. The conditional variance is constant in  $x$  for this Gaussian model. General conditional distributions need not have a linear mean or a constant variance.

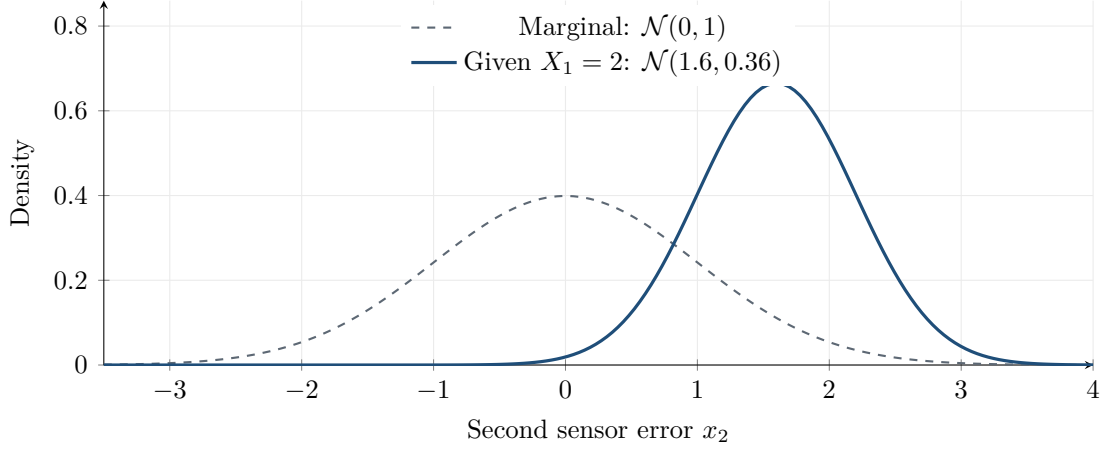


Figure 3.3: Before observing the first sensor, the second error has distribution  $\mathcal{N}(0, 1)$ . After observing  $X_1 = 2$ , it has distribution  $\mathcal{N}(1.6, 0.36)$ . Knowledge of the shared disturbance shifts the mean and reduces the remaining spread.

Total variance checks the result directly:

$$\underbrace{\sigma_Y^2}_{\text{total variance}} = \underbrace{\sigma_Y^2(1 - \rho^2)}_{\text{remaining after observing } X} + \underbrace{\rho^2 \sigma_Y^2}_{\text{variation of the conditional mean}}.$$

In the numerical model,  $1 = 0.36 + 0.64$ . If  $\rho = 0$ , observing  $X$  changes neither the mean nor the variance, consistent with Gaussian independence. As  $|\rho|$  approaches one, the remaining variance approaches zero. At  $|\rho| = 1$ , the relationship is exact and the conditional distribution is a point mass.

### 3.10 Gaussian class-conditional models

Suppose an object belongs to one of several classes, recorded by  $C \in \{1, \dots, K\}$ , and  $\mathbf{X}$  is its vector of measured features. A Gaussian class-conditional model assumes

$$\mathbb{P}(C = k) = \pi_k, \quad \mathbf{X} \mid C = k \sim \mathcal{N}_d(\boldsymbol{\mu}_k, \Sigma_k).$$

Each class has its own center and covariance. For positive-definite  $\Sigma_k$ , let  $f_k$  denote its Gaussian density. Bayes' theorem from Chapter 1 now gives

$$\mathbb{P}(C = k \mid \mathbf{X} = \mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_j \pi_j f_j(\mathbf{x})}. \quad (3.20)$$

For a continuous feature vector, this is a density-based conditional probability, just as in (3.5). The denominator is the feature density after the class label is ignored: a Gaussian mixture of the kind introduced in Section 2.9.

### 3.10.1 The relationship between features can identify the class

Consider two equally likely operating modes of a device. In mode  $C = +$ , its two sensor errors tend to move together; in mode  $C = -$ , they tend to move in opposite directions. Use

$$\boldsymbol{\mu}_+ = \boldsymbol{\mu}_- = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \Sigma_+ = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}, \quad \Sigma_- = \begin{pmatrix} 1 & -0.8 \\ -0.8 & 1 \end{pmatrix}.$$

Each feature alone has distribution  $\mathcal{N}(0, 1)$  in both classes. Its value alone therefore leaves the class probabilities at  $1/2$ . The distinguishing information is entirely in the pair.

The determinants are both 0.36, so the density normalization constants cancel in the likelihood ratio. Subtracting the exponents in (3.16) gives

$$\log \frac{f_+(x_1, x_2)}{f_-(x_1, x_2)} = \frac{2(0.8)}{1 - 0.8^2} x_1 x_2 = \frac{40}{9} x_1 x_2.$$

With equal prior probabilities, mode  $+$  is more probable when  $x_1 x_2 > 0$ , and mode  $-$  is more probable when  $x_1 x_2 < 0$ . At  $(1, 1)$ ,

$$\mathbb{P}(C = + \mid \mathbf{X} = (1, 1)) = \frac{1}{1 + e^{-40/9}} \approx 0.9884.$$

At  $(1, -1)$ , this probability is approximately 0.0116. The individual feature values have equal marginal densities in both classes; their agreement or disagreement changes the conclusion.

If we incorrectly replace each class density by the product of its feature marginals, both models become  $\phi(x_1)\phi(x_2)$ . Then every pair receives posterior class probability  $1/2$ . Assuming conditional independence has erased all the useful information in this example.

### 3.10.2 What the Gaussian decision rule uses

For positive class priors, maximizing the posterior is equivalent to maximizing the log-score

$$s_k(\mathbf{x}) = \log \pi_k - \frac{1}{2} \log \det \Sigma_k - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k).$$

The omitted term  $-\frac{d}{2} \log(2\pi)$  is the same for every class. The three remaining terms reflect the class's prevalence, the volume over which its density is spread, and the observation's displacement relative to the class's covariance. A broad class cannot receive the same peak density as a narrow class with the same total probability.

If all classes share a common covariance  $\Sigma$ , expanding the quadratic expression shows that the term  $-\frac{1}{2} \mathbf{x}^\top \Sigma^{-1} \mathbf{x}$  is also common. The class-dependent part becomes

$$\mathbf{x}^\top \Sigma^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^\top \Sigma^{-1} \boldsymbol{\mu}_k + \log \pi_k,$$

which is linear in the observed features, plus a constant. Comparing two classes therefore gives a linear boundary. If the covariance matrices differ, quadratic terms generally remain, as they did in the two-mode example.

These conclusions come from specifying a joint probability model and conditioning on an observation. How to learn the means, covariance matrices, and class probabilities from data belongs to statistical estimation. Here the important point is that a feature's usefulness can depend on its relationship with other features, not just its own distribution.